Meta RL

(and how it relates to Generalization)

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Disclaimer

- Meta RL and RL generalization are much broader than the specific narrative in this talk.
- For comprehensive coverage start here:
  - Ghavamzadeh, Mohammad, et al. "Bayesian reinforcement learning: A survey." *Foundations and Trends in Machine Learning* 8.5-6 (2015): 359-483. https://arxiv.org/abs/1609.04436
  - Beck, Jacob, et al. "A tutorial on meta-reinforcement learning." *Foundations and Trends in Machine Learning* 18.2-3 (2025): 224-384. https://arxiv.org/abs/2301.08028
  - Kirk, Robert, et al. "A survey of zero-shot generalisation in deep reinforcement learning." *Journal of Artificial Intelligence Research* 76 (2023): 201-264. https://arxiv.org/abs/2111.09794

Where Are The Robots?

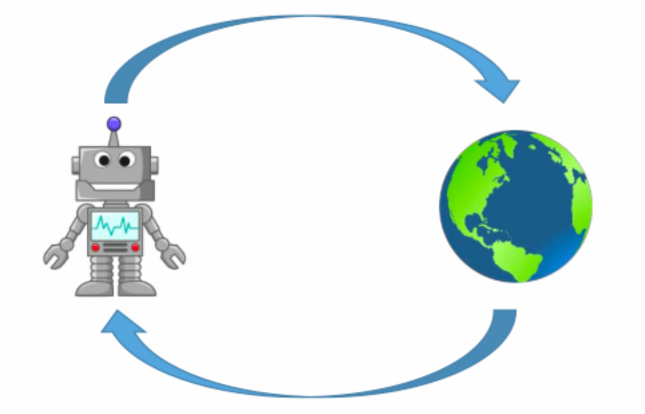
- Extreme **uncertainty**
- Must **generalize, adapt**
- Reinforcement learning?





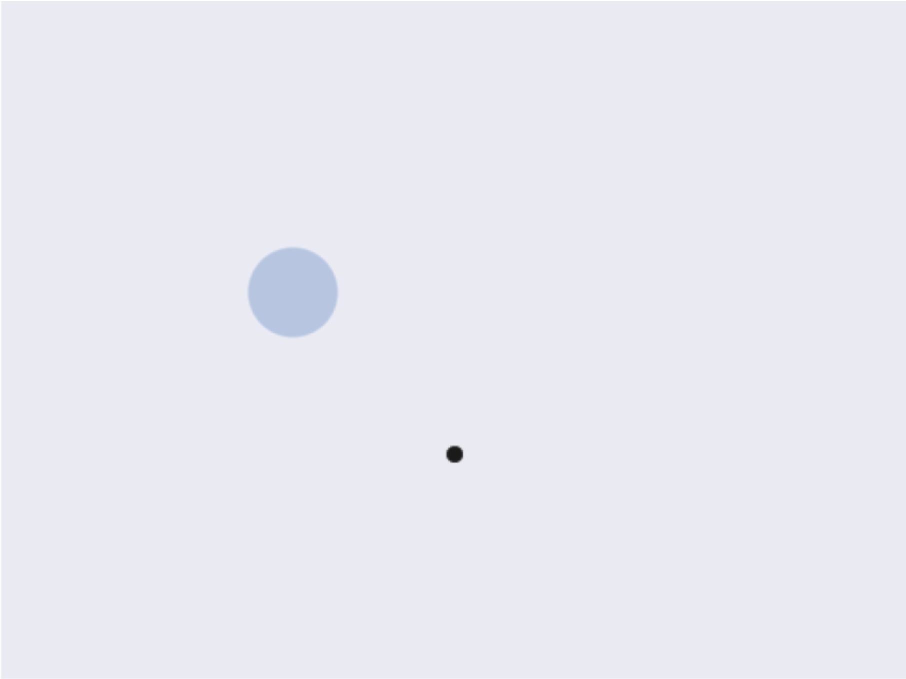

Motivation

- The deep RL promise



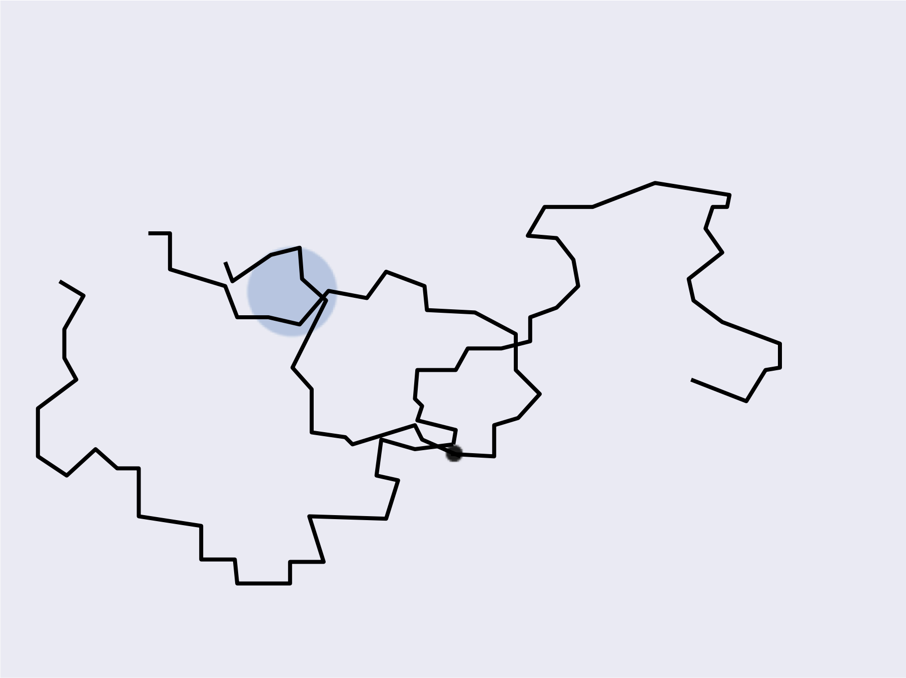
Learning decision making from interaction

Is RL Enough?

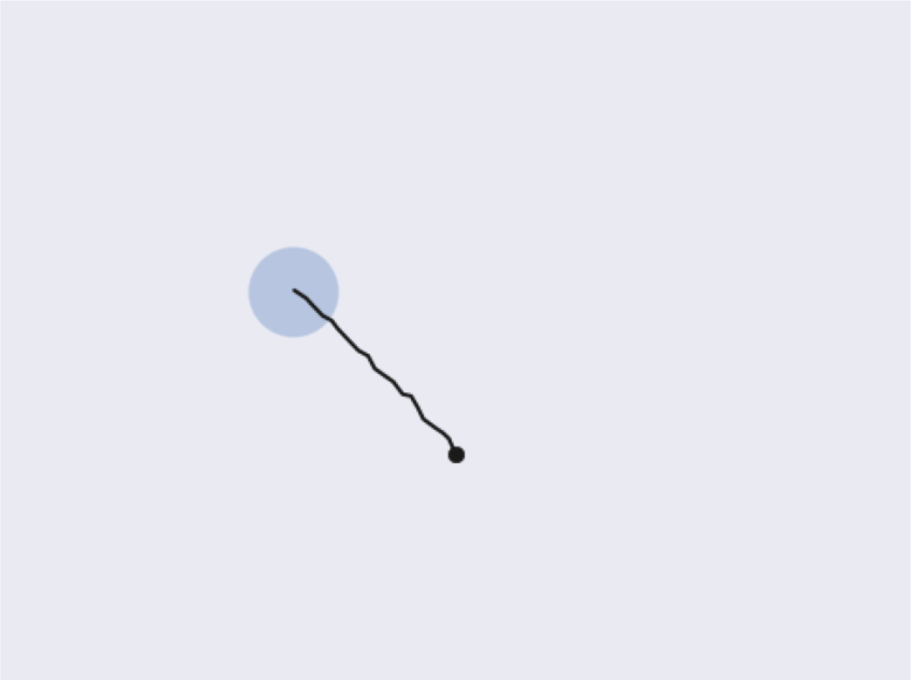


How to reach the goal?

Is RL Enough?

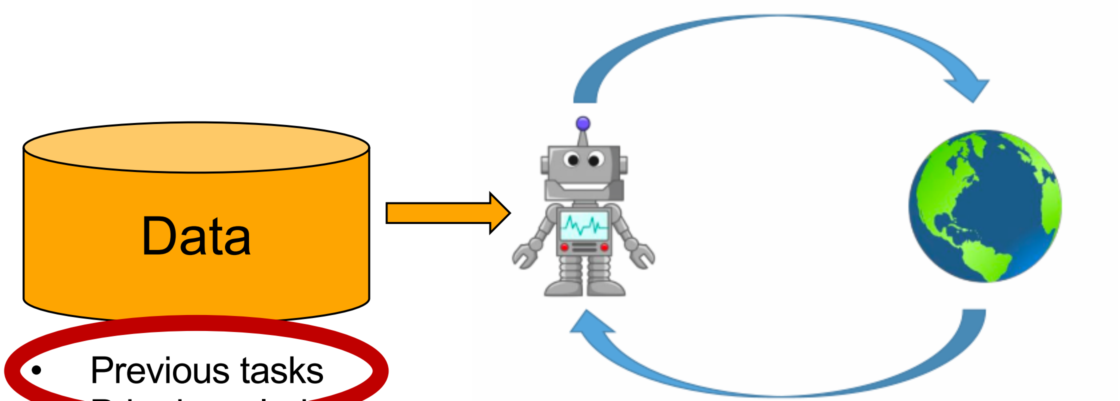


Is RL Enough?



Fast RL

- The deep RL promise



- Previous tasks
- Prior knowledge
- Other agents
- Demonstrations
- World models
- LLMs
- …

Meta RL [1-3]

Lifelong learning [4]

Transfer learning [5]

Bayesian RL [6]

Generalization in RL [7]

…

[1] Duan et al., RL^2 , 2016
[2] Finn et al., MAML, 2017
[3] Dorfman et al., BOReL, 2021
[4] Thrun and Mitchell, Lifelong robot learning, 1995
[5] Taylor and Stone, Transfer learning for RL domains: a survey, 2009
[6] Ghavamzadeh et al., Bayesian RL: a survey, 2016
[7] Cobbe et al., Procgen benchmark, 2020

Fundamental Question

What can we learn from training tasks?

Supervised learning
Training tasks → decision rule

Reinforcement learning
Training tasks → ?

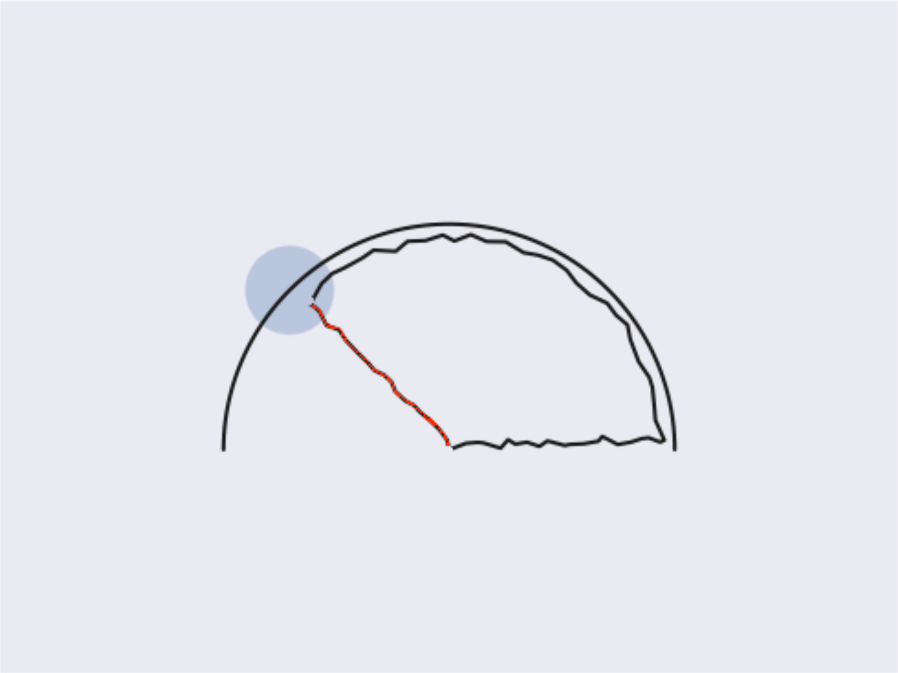
Learning from previous tasks

- Goal location is unknown
- **Data** from agents trained to reach different goals

Learning from previous tasks

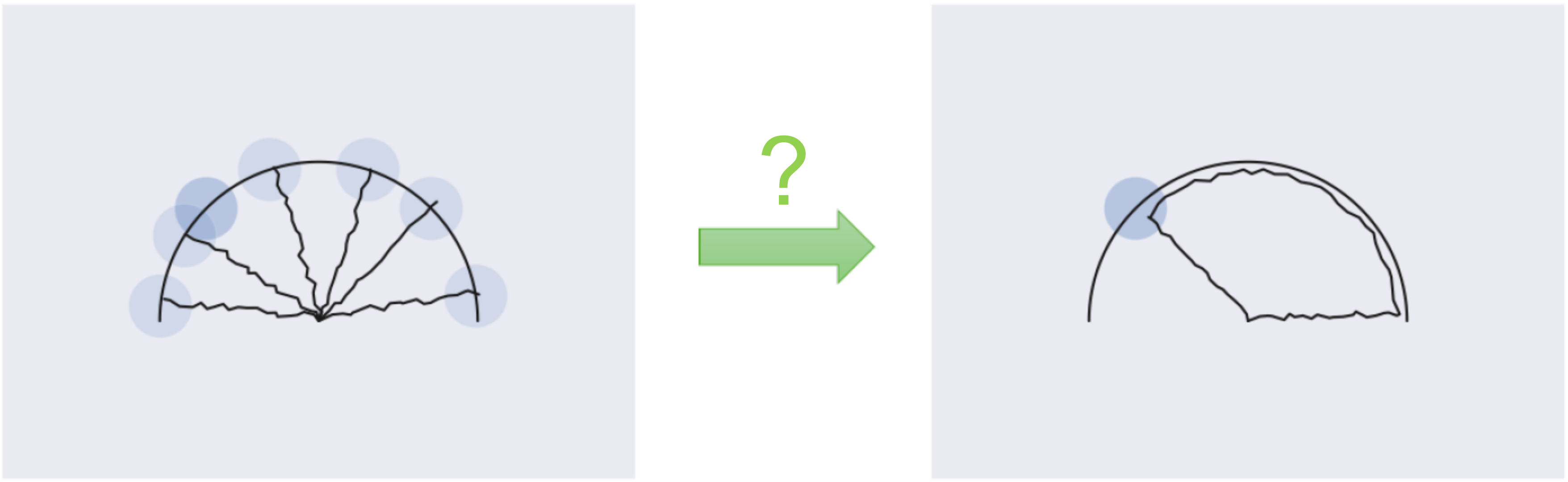
1. Search optimally across semi-circle

Learning from previous tasks



1. Search optimally across semi-circle
2. Go to found goal

Learning from previous tasks



Can we use **data** to learn to **explore**?

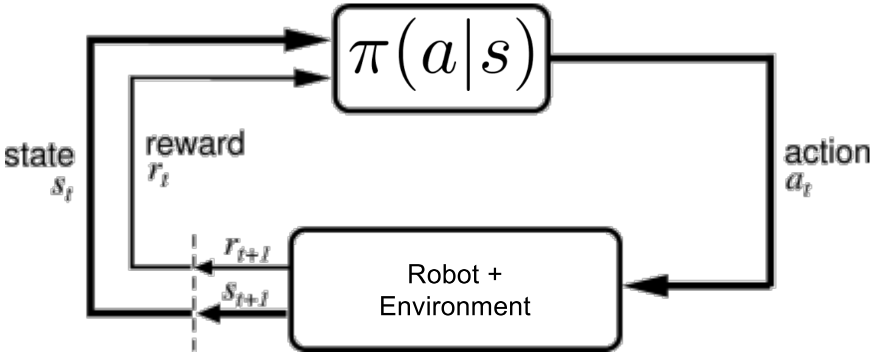
Fundamental Questions

1. What is **optimal exploration**?
2. How much **data** do we need to **generalize**?
3. Effective **algorithms**?

Reinforcement Learning

Markov Decision Process (MDP)

- States s, actions a
- Transitions $P(s'|s,a)$
- Policy $\pi(a|s)$
- Reward $r(s)$



Goal: $$\max_{\pi} \mathbb{E}\left[\sum_{t=0}^{T} r_t\right]$$

Exploration in RL

No prior information

- UCRL
- E^3
- Exploration bonuses
- ϵ-greedy
- …
- ‘don’t waste time on non-interesting states’

Regret bounds, PAC bounds, …

Prior over MDP

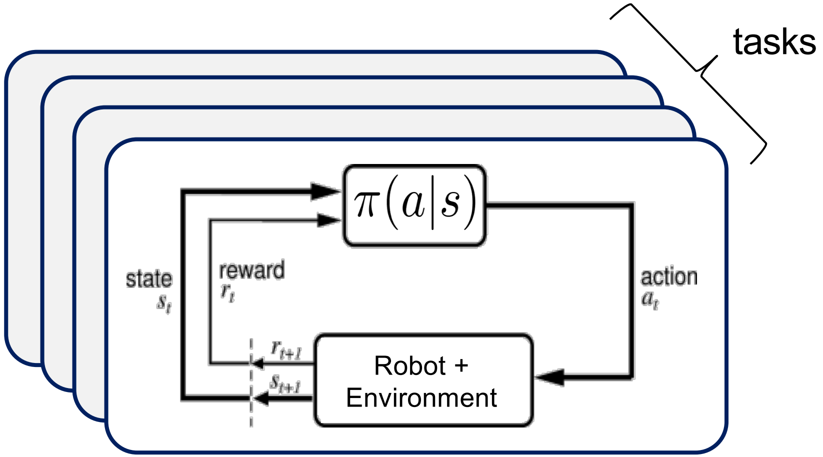
- Bayesian RL
- ‘plan to obtain information/reward’

Well-defined **optimal** exploration!

Bayesian RL (a.k.a. Meta RL)

- Multi-task RL
- Each task MDP $M = \{P(s'|s,a), r(s)\}$
- Task distribution $P(M)$

Goal: $\max_{\pi} \mathbb{E}_{M \sim P(M)} \left[\mathbb{E}_M \left[\sum_{t=0}^{H} r_t \right] \right]$



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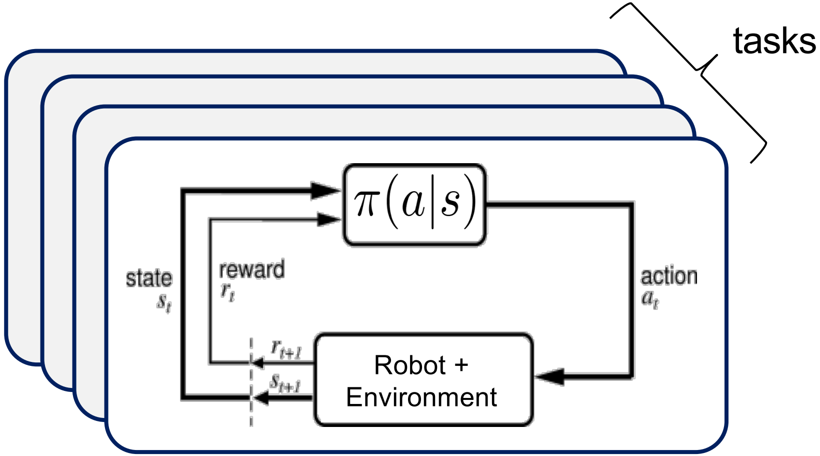

Goal: $\max_{\pi} \mathbb{E}_{M\sim P(M)}\left[\mathbb{E}_M\left[\sum_{t=0}^{H} r_t\right]\right]$

Non-Markov policy - optimally **explore/exploit!**

Agent is “thrown” into **unknown** task

Agent has H steps to max reward

This objective only LOOKS simple!

Bayesian RL (a.k.a. Meta RL)

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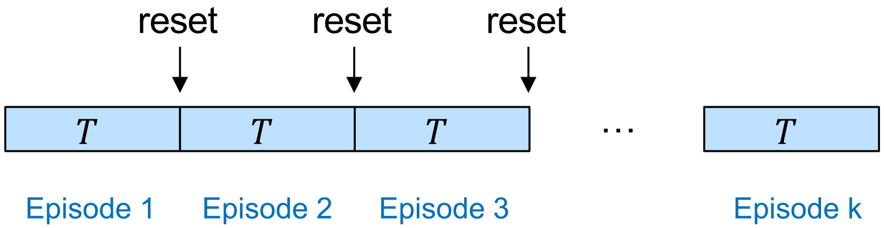
Goal: $$\max_{\pi} \mathbb{E}_{M\sim P(M)}\left[\mathbb{E}_M\left[\sum_{t=0}^{H=kT} r_t\right]\right]$$

k-shot learning variant



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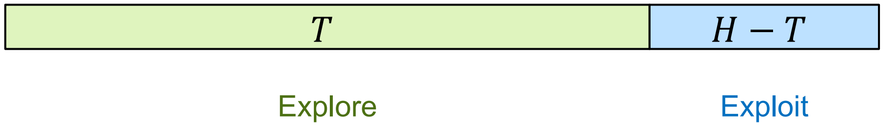
Explore-then-exploit variant



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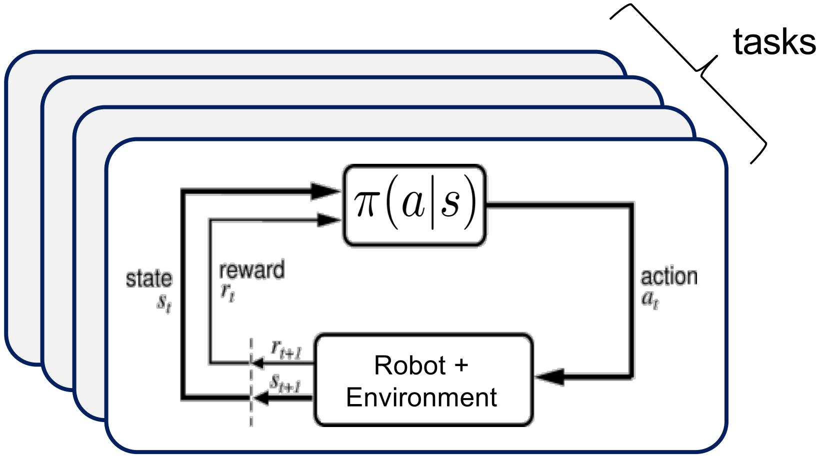
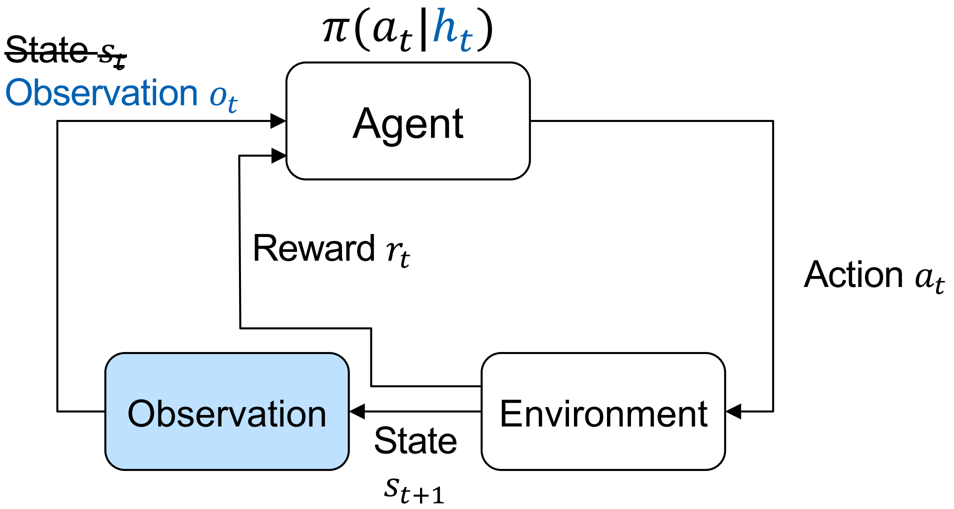


- **Special case of partially observed MDP – POMDP**
- **We should study POMDPs**

Partially Observable MDP

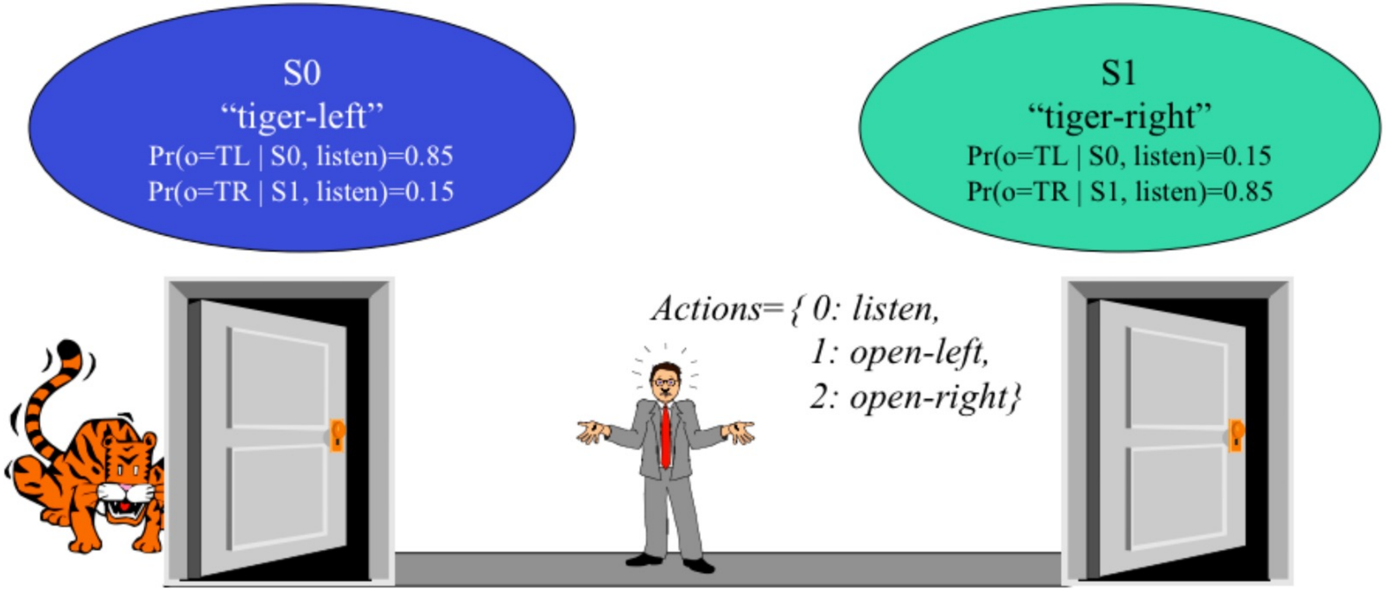
- States s, actions a
- Transitions $P(s'|s,a)$
- Observations $P(o|s)$
- History $h_t = o_0, a_0, r_0 \dots, o_t$
- Policy $\pi(a_t|h_t)$
- Reward $r(s)$



Goal: $$\max_{\pi} \mathbb{E}\left[\sum_{t=0}^{T} r_t\right]$$

Partially Observable MDP

- Example:



Reward Function

- *Penalty for wrong opening: -100*
- *Reward for correct opening: +10*
- *Cost for listening action: -1*

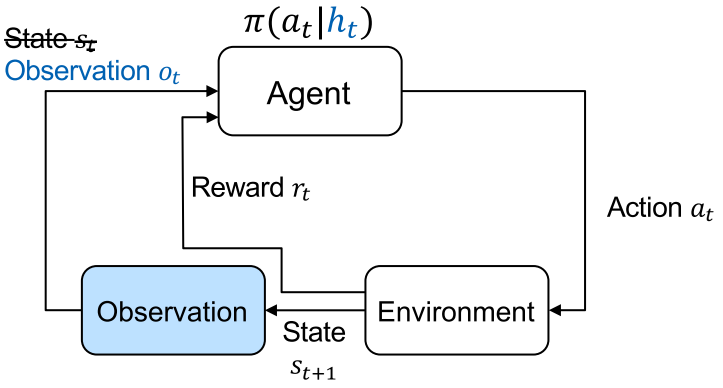
Observations

- *to hear the tiger on the left (TL)*
- *to hear the tiger on the right(TR)*

Slide credits:
Pieter Abbeel
Hui Huang

Partially Observable MDP

- Observations $P(o|s)$
- History $h_t = o_0, a_0, r_0 \dots, o_t$
- Policy $\pi(a_t|h_t)$



Good news:

- POMDP = MDP over histories
  - $h_{t+1} = \{h_t, a_t, r_t, o_{t+1}\}$
  - $\bar{P}(h_{t+1}|h_t, a_t, r_t)$

Underlying MDP

$$= \sum_{s_t} P(s_t|h_t, r_t) \sum_{s'} P(s'|s_t, a_t) P(o_{t+1}|s')$$

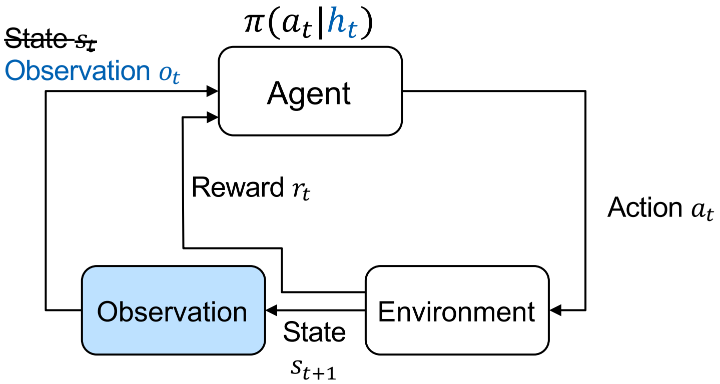
Well defined (Bayes rule)

Observations $P(o|s)$

Partially Observable MDP

Dynamic Programming

- $V_t^\pi(h_t) = \mathbb{E}^\pi[\sum_{s=t}^T r_s | h_t]$
- $V_t^\pi(h_t) = \mathbb{E}^\pi[r_t + V_{t+1}^\pi(h_{t+1}) | h_t]$
- $V_t^*(h_t) = \max_{a_t} \mathbb{E}[r_t + V_{t+1}^*(h_{t+1}) | h_t, a_t]$



Bad news:

- History space grows **exponentially** with H

Partially Observable MDP

Information States[1]

- $z_t = f_t(h_t)$

Predict reward

1. $\mathbb{E}[r_t|h_t, a_t] = \mathbb{E}[r_t|z_t, a_t]$
2. $P[z_{t+1}|h_t, a_t] = P[z_{t+1}|z_t, a_t]$

Predict itself

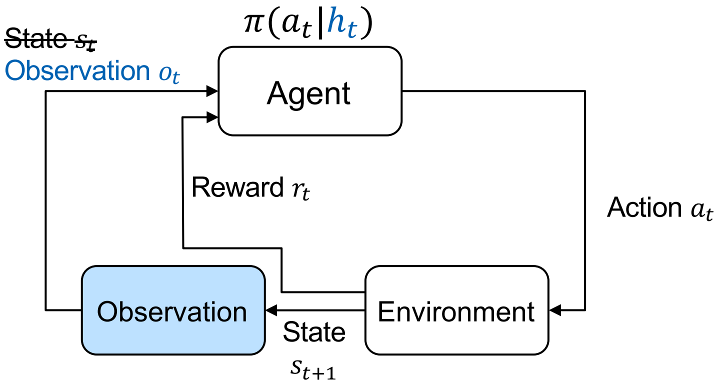
Alt. (stricter)

1. $z_{t+1} = g_t\,(z_t, o_t, a_t)$

State evolution

2. $P[o_{t+1}|h_t, a_t] = P[o_{t+1}|z_t, a_t]$

Predict next observation



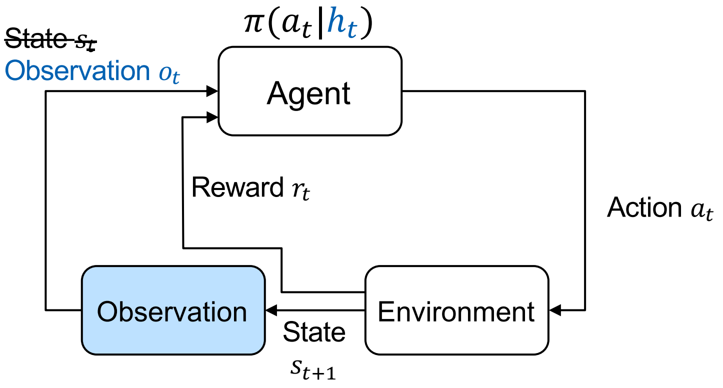
[1] Subramanian et al. Approximate information state for approximate planning and RL in partially observed systems. JMLR 2022

Partially Observable MDP

Dynamic Programming

- $\bar{Q}_t^*(z_t, a_t) = \mathbb{E}[r_t + \bar{V}_{t+1}^*(z_{t+1}) | z_t, a_t]$
- $\bar{V}_t^*(z_t) = \max_{a_t} \bar{Q}_t^*(z_t, a_t)$

Thm.
- $Q_t^*(h_t, a_t) = \bar{Q}_t^*(z_t, a_t)$
- $V_t^*(h_t) = \bar{V}_t^*(z_t)$



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Partially Observable MDP

Dynamic Programming

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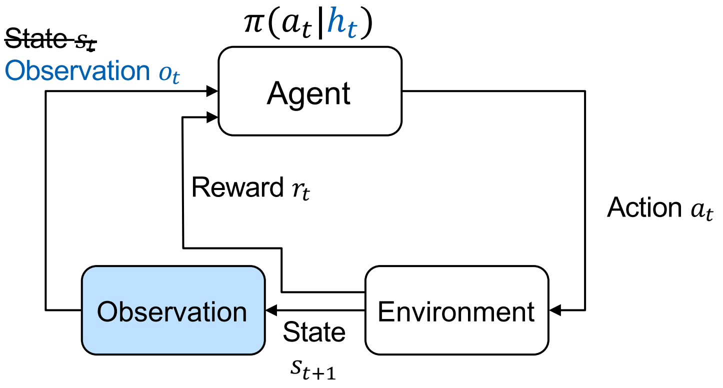

Thm.

- $Q_t^*(h_t, a_t) = \bar{Q}_t^*(z_t, a_t)$
- $V_t^*(h_t) = \bar{V}_t^*(z_t)$

Info. State predicts reward, itself

Induction assumption

Proof by induction:

- $Q_t^*(h_t, a_t) = \mathbb{E}[r_t + V_{t+1}^*(h_{t+1}) | h_t, a_t] = \mathbb{E}[r_t + \bar{V}_{t+1}^*(z_{t+1}) | h_t, a_t]$
  $= \mathbb{E}[r_t + \bar{V}_{t+1}^*(z_{t+1}) | z_t, a_t] = \bar{Q}_t^*(z_t, a_t)$

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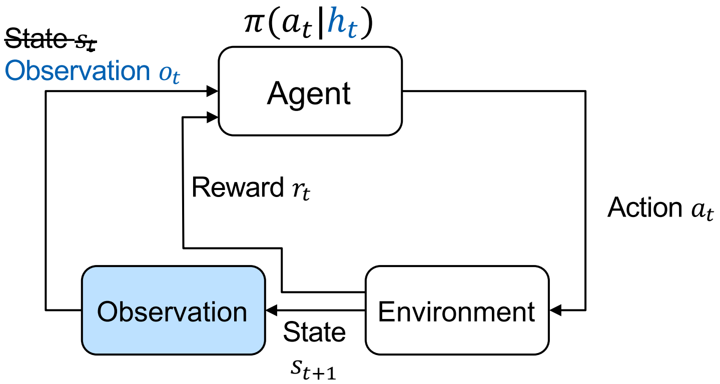
Partially Observable MDP

Information States Examples

- *(obvious)* $z_t = h_t$
- *(belief)* $z_t = P(s_t | h_t)$

Main insight: latent world models!



Information States = World Models!

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Predict reward

Predict itself

Alt. (stricter)

1. $z_{t+1} = g_t(z_t, o_t, a_t)$
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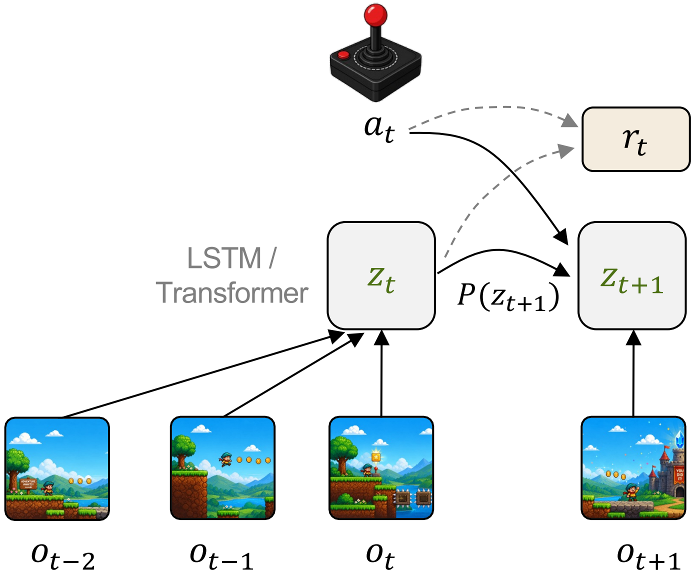
State evolution

Predict next observation

World Models



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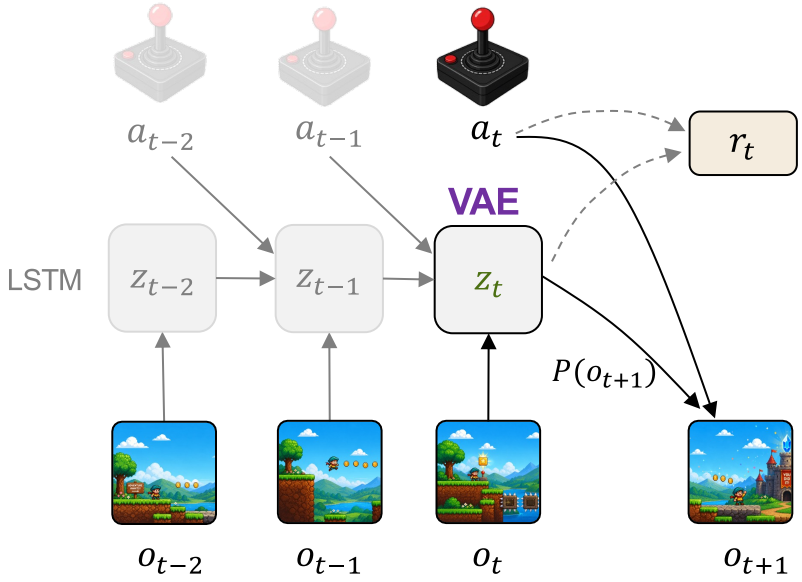
Predict next observation

World Models



Zintgraf et al. **VariBAD**: A Very Good Method for Bayes-Adaptive Deep RL via Meta-Learning. 2019

Information States = World Models!

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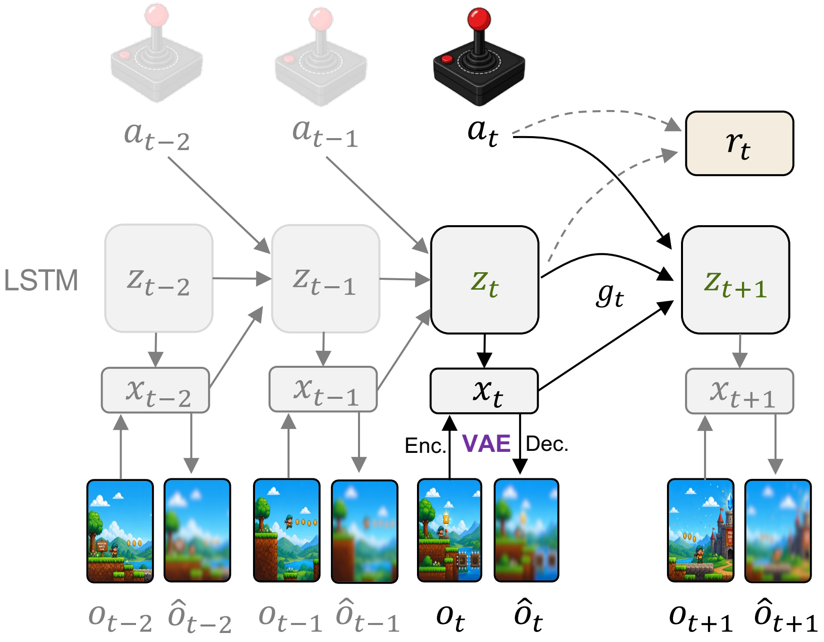
World Models



Dreamer V3. Danijar, et al. Mastering diverse control tasks through world models. Nature 2025.

Information States = World Models!

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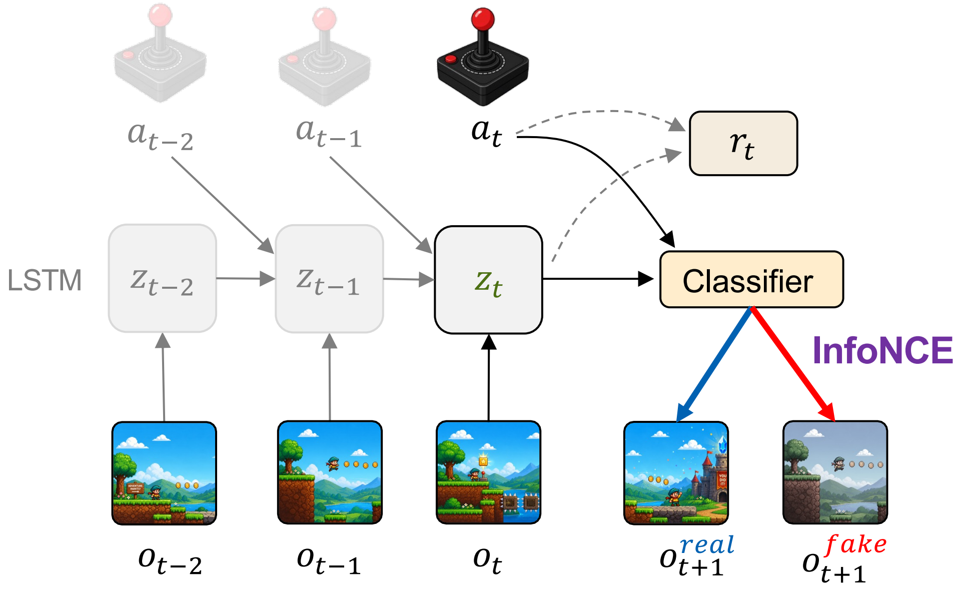
Predict next observation

World Models



Choshen, **T.** **ContraBAR**: Contrastive Bayes-Adaptive Deep RL. ICML 2023.

Information States = World Models!

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Predict reward

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State evolution

Predict next observation

World Models

Extensive **self-supervised learning** techniques

- (Variational) autoencoders
- InfoNCE (e.g., CPC, CLIP)
- BYOL (e.g., V-JEPA)
- Many others[1]

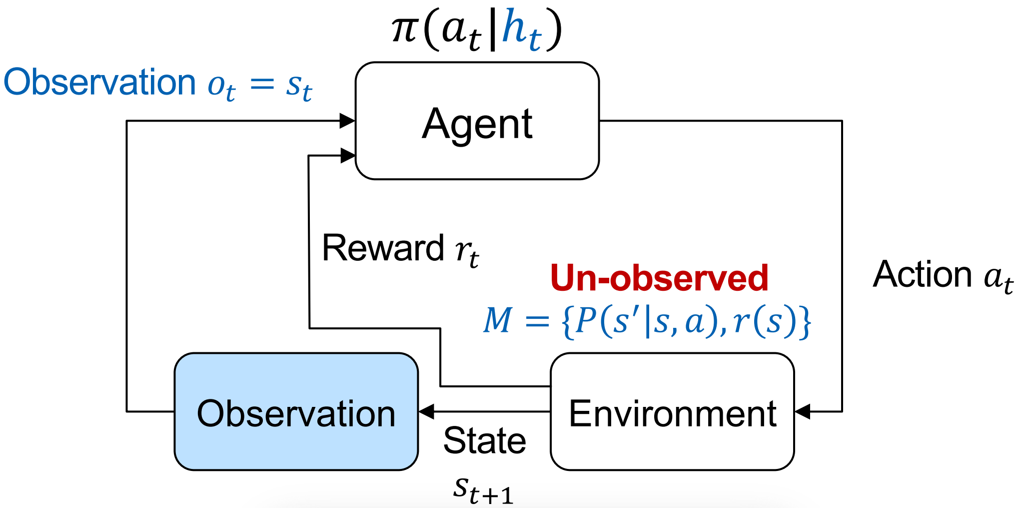
[1] Gui, Jie, et al. "A survey on self-supervised learning: Algorithms, applications, and future trends." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 46.12 (2024): 9052-9071.

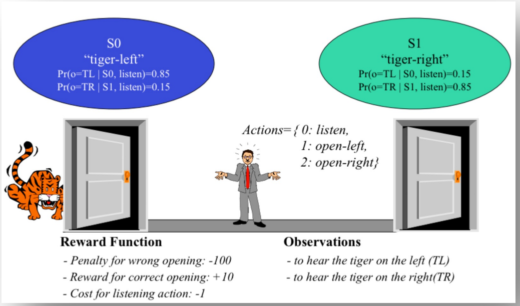
Back to Meta RL

$$\mathbb{E}_{M\sim P(M)}\left[\mathbb{E}_M\left[\sum_{t=0}^{H} r_t\right]\right]$$

Task distribution

Performance in specific task





A. Tamar, Technion

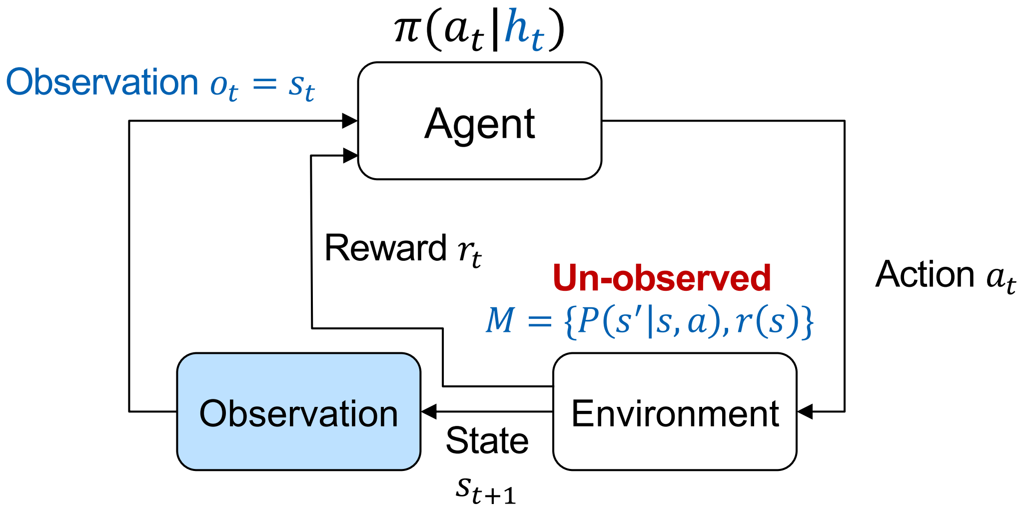
Back to Meta RL

$$\overset{\frac{1}{N}\sum_{i=1}^{N}}{\cancel{\mathbb{E}_{M\sim P(M)}}}\left[\mathbb{E}_{M_i}\left[\sum_{t=0}^{H} r_t\right]\right]$$

Training/test tasks

$M_i \sim P(M)$

Performance in specific task



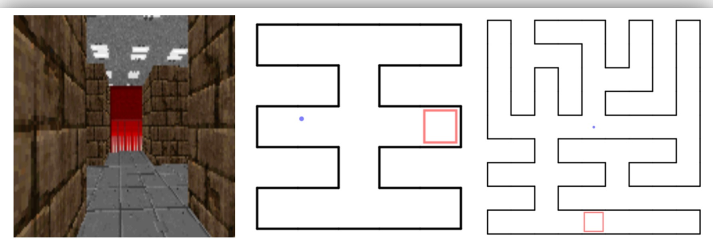
For now: training tasks = test tasks (later: generalization)

Meta RL Algorithms

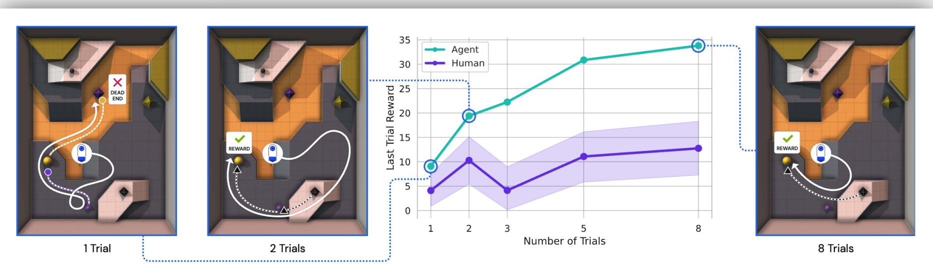
RL² – the Simplest Meta RL Algorithm

- Directly map history → action using LSTM/transformer
- Train with online RL (e.g., TRPO, PPO, etc.)

$h_t \longrightarrow$ **NN Policy** $\longrightarrow a_t$



[1]

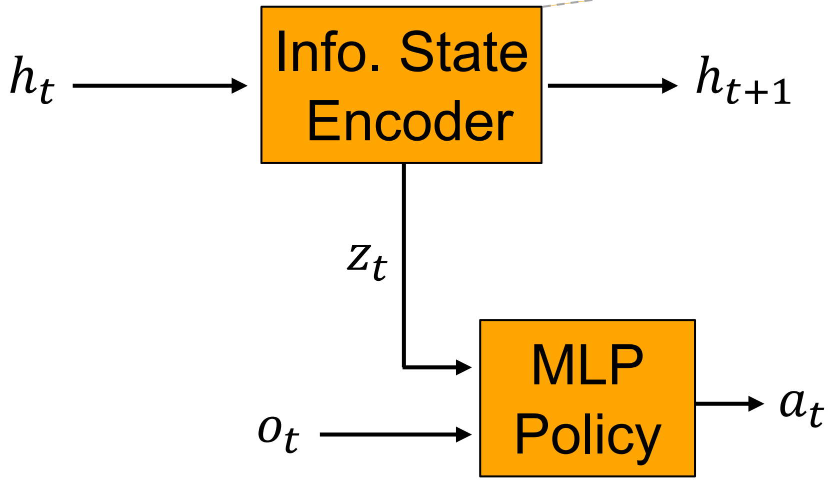


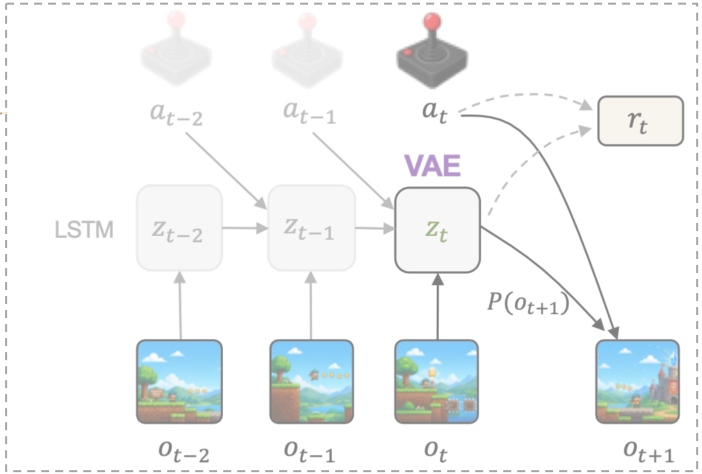
[2]

[1] Duan et al., RL²: Fast Reinforcement Learning via Slow Reinforcement Learning, 2016.
[2] Deepmind Adaptive Agent Team, et al. Human-timescale adaptation in an open-ended task space. 2023.

VariBAD

- Learn to map history → information state
- Train policy with online RL (e.g., PPO, SAC)



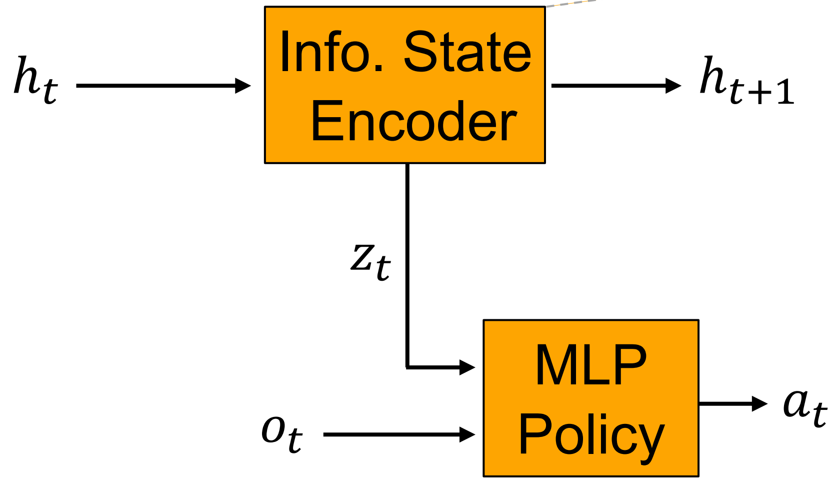


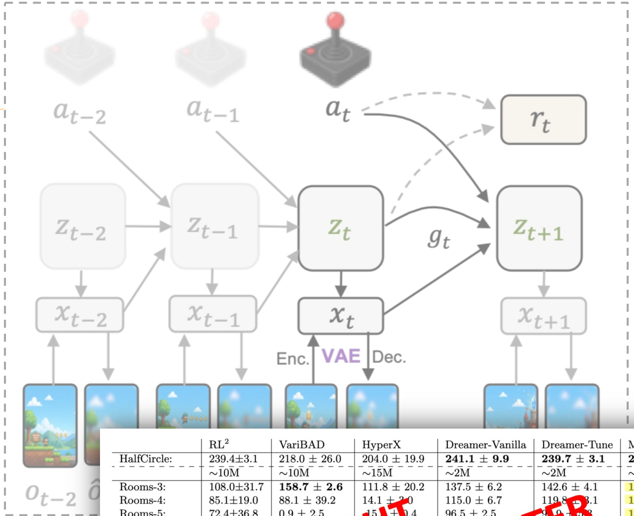
[1]

[1] Zintgraf et al. VariBAD: A Very Good Method for Bayes-Adaptive Deep RL via Meta-Learning. 2019

Dreamer / MAMBA

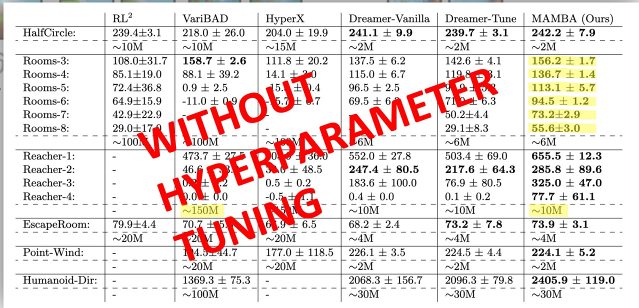
- Learn to map history → information state
- Train policy with Dreamer[1] (model based reinforce)
  + meta-RL specific tricks[2]





[1]

[2]



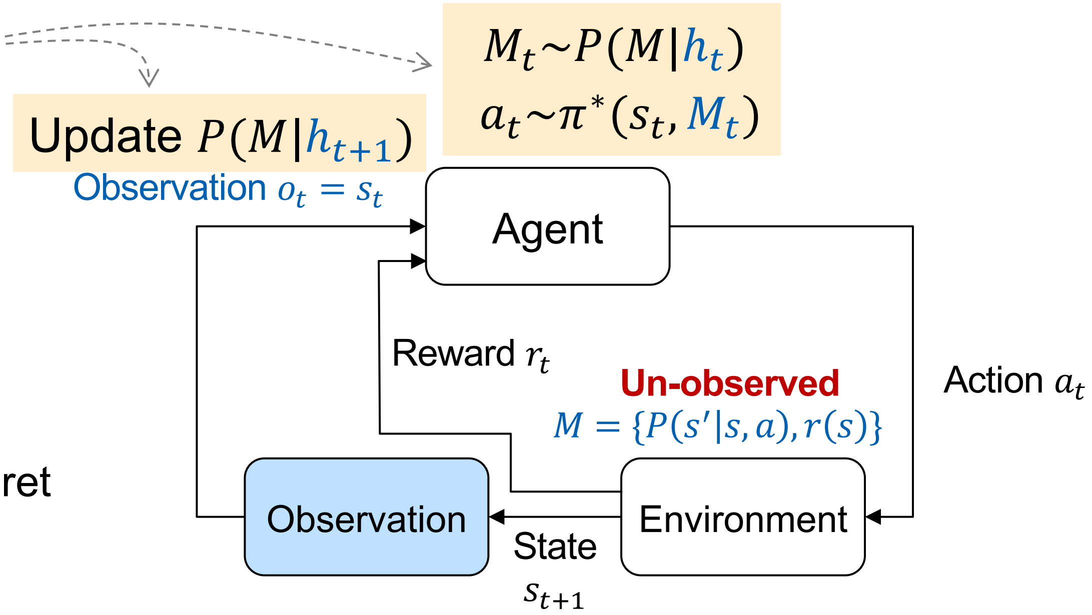
	RL^2	VariBAD	HyperX	Dreamer-Vanilla	Dreamer-Tune	MAMBA (Ours)
HalfCircle:	239.4±3.1	218.0 ± 26.0	204.0 ± 19.9	**241.1 ± 9.9**	**239.7 ± 3.1**	**242.2 ± 7.9**
	∼10M	∼10M	∼15M	∼2M	∼2M	∼2M
Rooms-3:	108.0±31.7	**158.7 ± 2.6**	111.8 ± 20.2	137.5 ± 6.2	142.6 ± 4.1	156.2 ± 1.7
Rooms-4:	85.1±19.0	88.1 ± 39.2	14.1 ± [illegible]	115.0 ± 6.7	119.8 ± [illegible]	136.7 ± 1.4
Rooms-5:	72.4±36.8	0.9 ± 2.5	[illegible]	96.5 ± 2.5	[illegible]	113.1 ± 5.7
Rooms-6:	64.9±15.9	-11.0 ± 0.9	[illegible]	69.5 ± [illegible]	[illegible] ± 6.3	94.5 ± 1.2
Rooms-7:	42.9±22.9	-	-	[illegible]	50.2±4.4	73.2±2.9
Rooms-8:	29.0±17.0	[illegible]	-	[illegible]	29.1±8.3	55.6±3.0
	∼100M	∼100M	[illegible]	∼6M	∼6M	∼6M
Reacher-1:	-	473.7 ± 27.5	[illegible] ± 36.0	552.0 ± 27.8	503.4 ± 69.0	**655.5 ± 12.3**
Reacher-2:	-	46.6 ± [illegible]	[illegible] ± 48.5	**247.4 ± 80.5**	**217.6 ± 64.3**	**285.8 ± 89.6**
Reacher-3:	-	[illegible]	0.5 ± 0.2	183.6 ± 100.0	76.9 ± 80.5	**325.0 ± 47.0**
Reacher-4:	-	0.0 ± 0.0	-0.5 ± 1.1	0.4 ± 0.0	0.1 ± 0.2	**77.7 ± 61.1**
	-	∼150M	[illegible]	∼10M	∼10M	∼10M
EscapeRoom:	79.9±4.4	70.7 ± [illegible]	[illegible] ± 6.5	68.2 ± 2.4	**73.2 ± 7.8**	**73.9 ± 3.1**
	∼20M	∼20M	∼20M	∼4M	∼4M	∼4M
Point-Wind:	-	[illegible]±44.7	177.0 ± 118.5	226.1 ± 3.5	224.5 ± 4.4	**224.1 ± 5.2**
	-	∼20M	∼20M	∼2M	∼2M	∼2M
Humanoid-Dir:	-	1369.3 ± 75.3	-	2068.3 ± 156.7	2096.3 ± 79.8	**2405.9 ± 119.0**
	-	∼100M	-	∼30M	∼30M	∼30M

WITHOUT HYPERPARAMETER TUNING

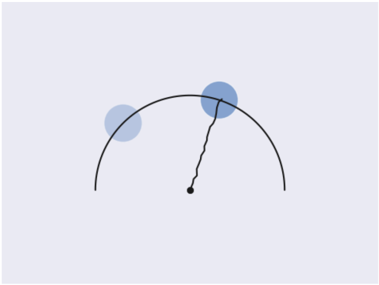
[1] Danijar, et al. Mastering diverse control tasks through world models. Nature 2025
[2] Rimon, Jurgenson, Krupnik, Adler, **T.** Mamba: an effective world model approach for meta-reinforcement learning. 2024

PEARL (Context-Based Meta RL)

Thompson Sampling

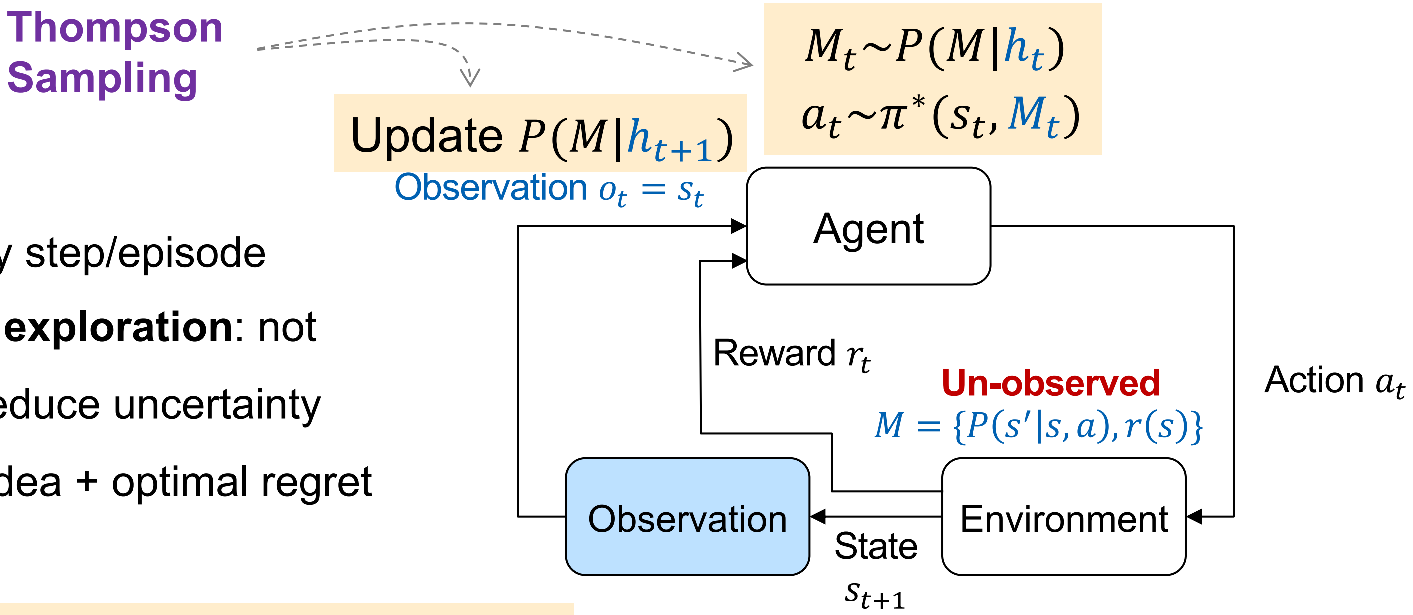


- Update every step/episode
- **Suboptimal exploration**: not directed to reduce uncertainty
- But, simple idea + optimal regret



[1] Rakelly et al. Efficient off-policy meta-reinforcement learning via probabilistic context variables. 2019

PEARL (Context-Based Meta RL)



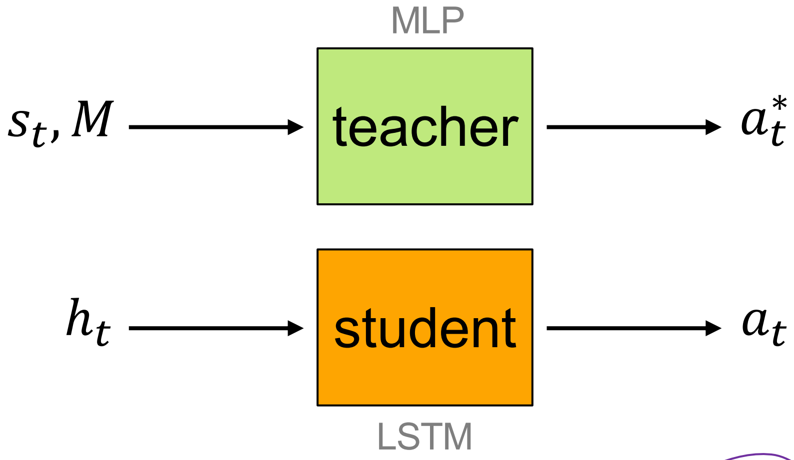
- Update every step/episode
- **Suboptimal exploration**: not directed to reduce uncertainty
- But, simple idea + optimal regret

- Train context-conditioned policy $\pi(s, z)$
- Train inference $P(z|h_t)$ using world model + tricks[1]
- Update every episode

[1] Rakelly et al. Efficient off-policy meta-reinforcement learning via probabilistic context variables. 2019

Teacher-Student

- Teacher observes the task
- Train teacher with online RL (e.g., TRPO, PPO, etc.)



Fully observed!

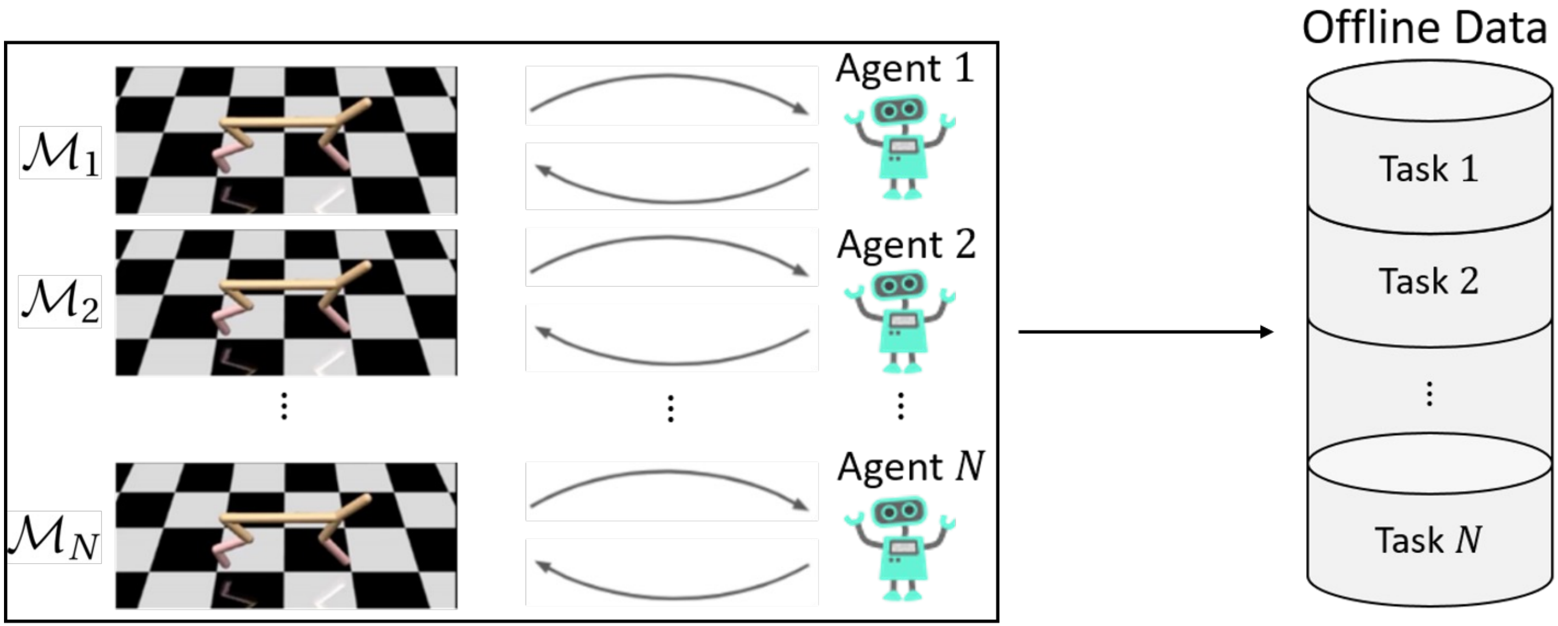
- Train student with: $(1-\lambda)L_{imitation} + \lambda \cdot L_{RL}$

$\lambda = 1 \rightarrow$ Thompson sampling
Automatically tune λ [1]

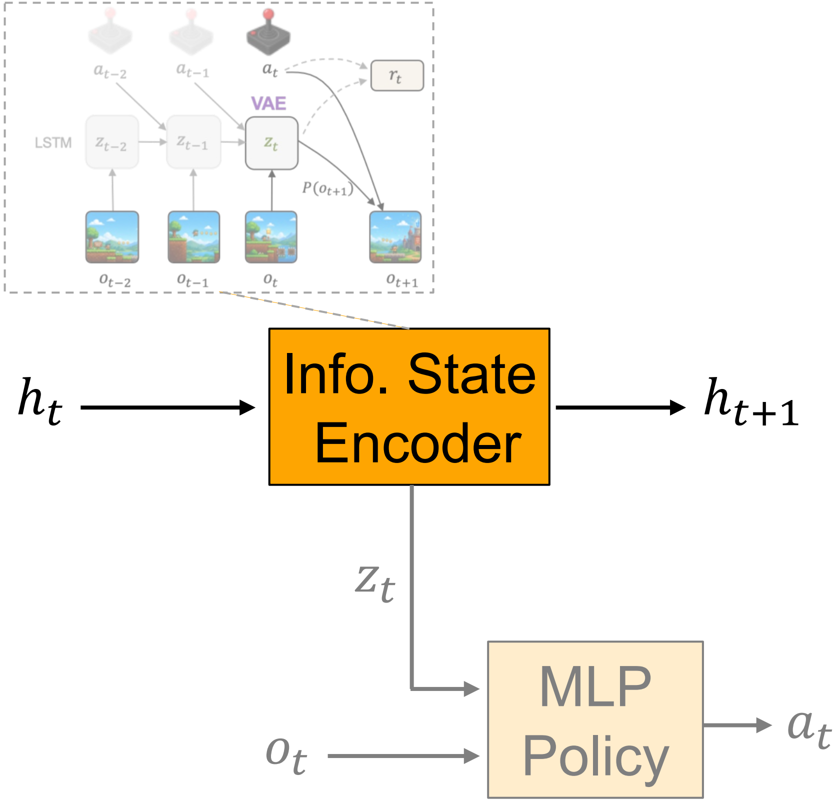
[1] Shenfeld, Hong, **T.**, Agrawal. TGRL: An Algorithm for Teacher Guided Reinforcement Learning, 2023

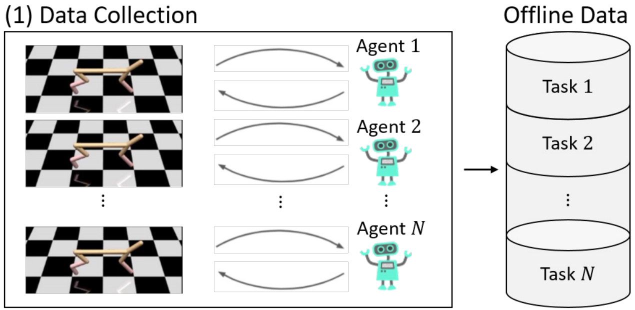
Can Meta RL Work Offline?

- Offline data - **entire** training histories of RL agents



Bayesian Offline Reinforcement Learning (BOReL)



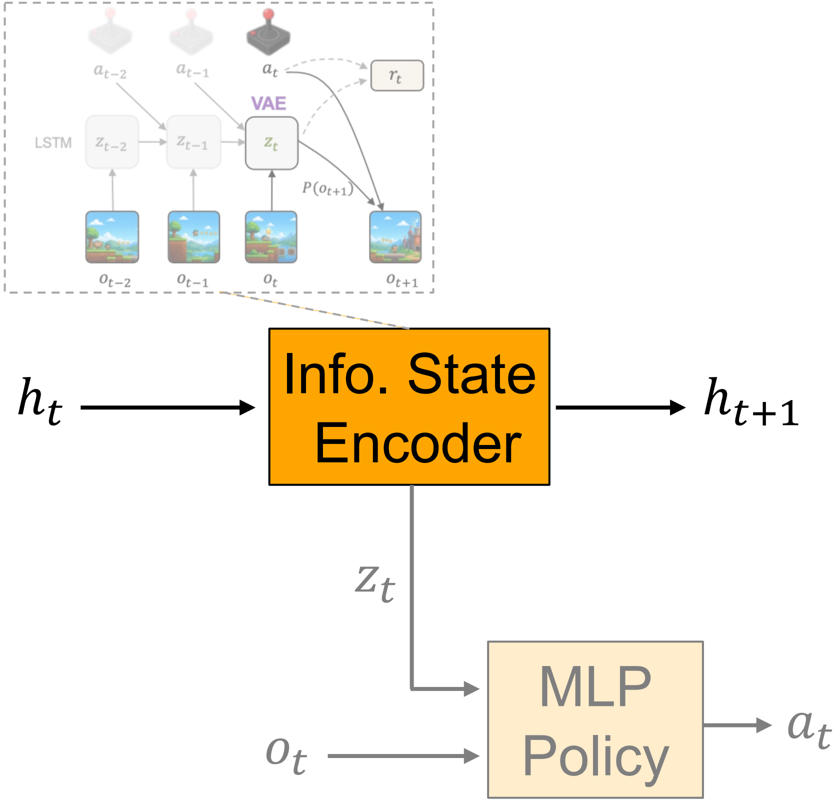


k episodes, agent j task j → h_t

$o_0^1, a_0^1, .. o_T^1$	$o_0^2, a_0^2, .. o_T^2$	$o_0^3, a_0^3, .. o_T^3$	$\cdots$	$o_0^k, a_0^k, .. o_T^k$
Episode 1	Episode 2	Episode 3		Episode k

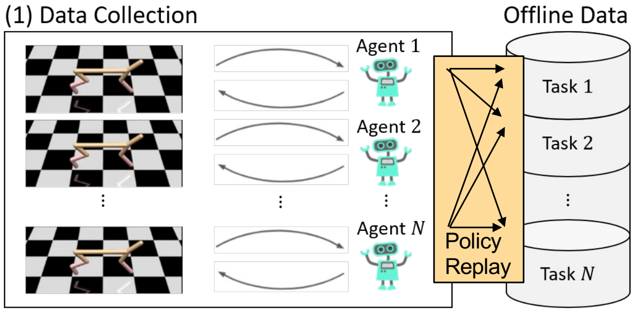
Dorfman, Shenfeld, **T.** Offline Meta RL – Identifiability Challenges and Effective Data Collection Strategies. NeurIPS 2021

Bayesian Offline Reinforcement Learning (BOReL)



Problem: MDP ambiguity[1]

Solution: policy replay

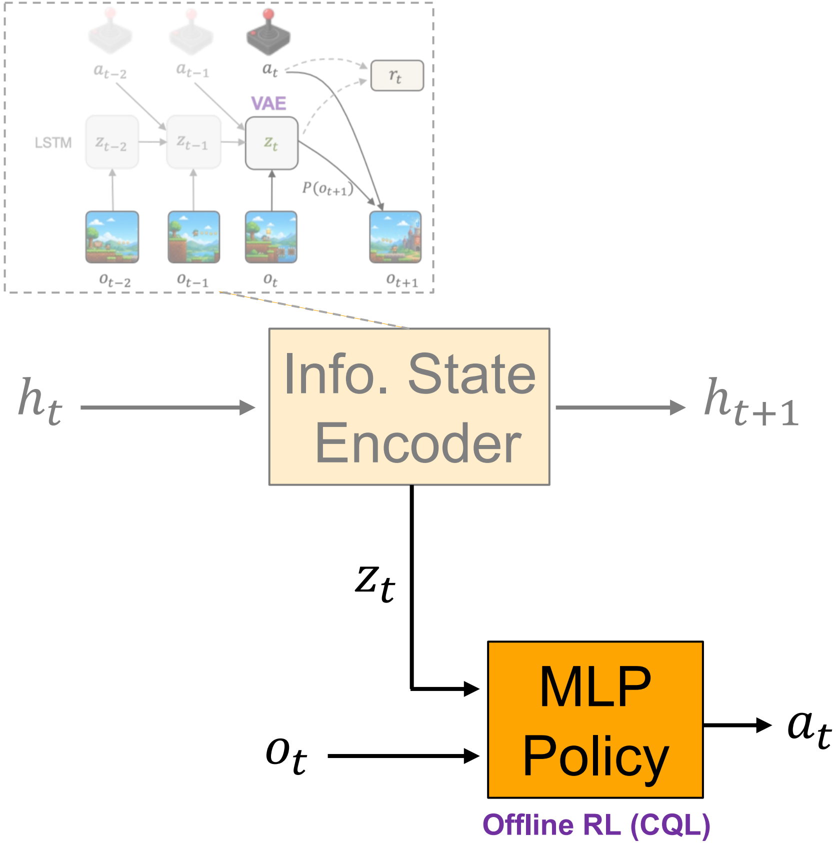


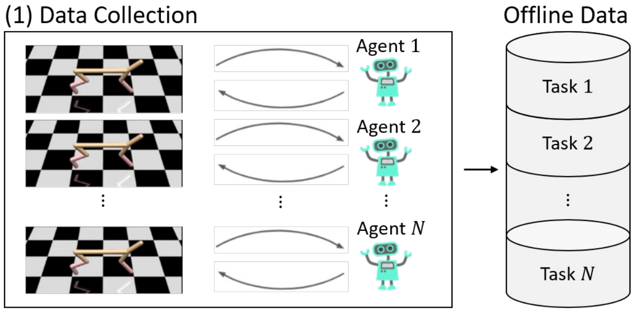
k episodes, ~~agent j~~ task j → h_t

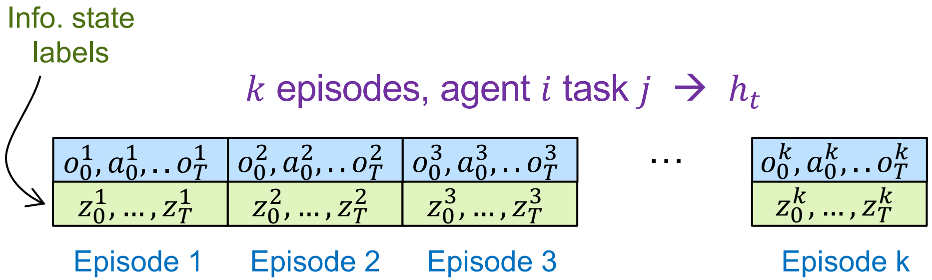
$o_0^1, a_0^1, .. o_T^1$	$o_0^2, a_0^2, .. o_T^2$	$o_0^3, a_0^3, .. o_T^3$	…	$o_0^k, a_0^k, .. o_T^k$
agent i	agent i'	agent i''		agent j
Episode 1	Episode 2	Episode 3		Episode k

[1] Dorfman, Shenfeld, **T.** Offline Meta RL – Identifiability Challenges and Effective Data Collection Strategies. NeurIPS 2021

Bayesian Offline Reinforcement Learning (BOReL)

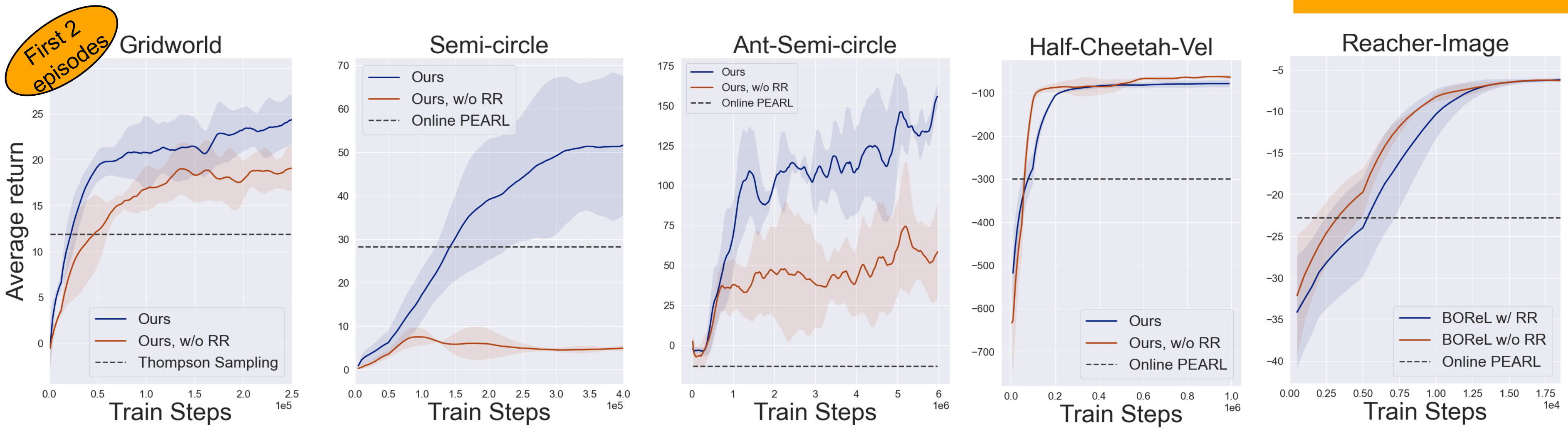


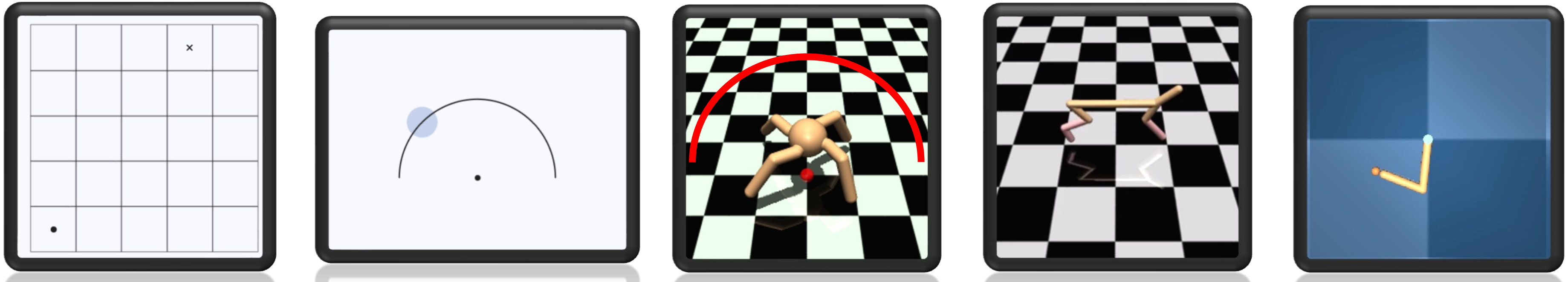




Dorfman, Shenfeld, **T.** Offline Meta RL – Identifiability Challenges and Effective Data Collection Strategies. NeurIPS 2021

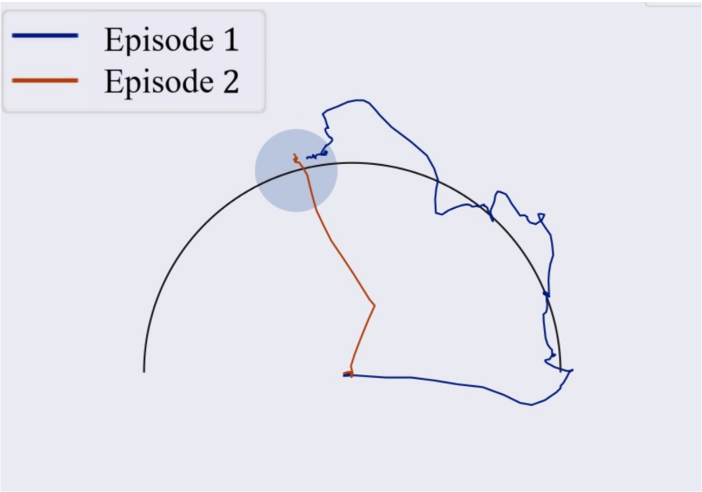
BOReL - Results

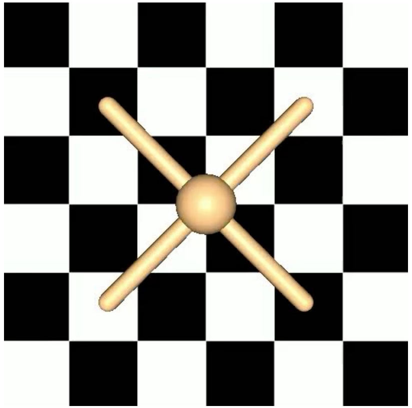


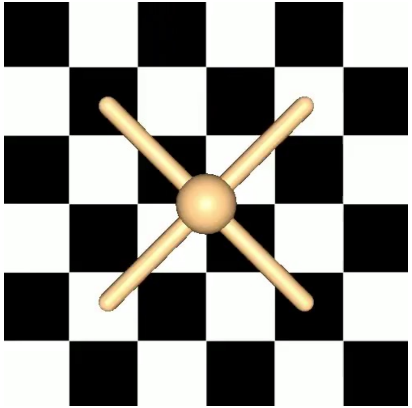


PEARL: Rakelly et al. Efficient off-policy meta-reinforcement learning via probabilistic context variables, 2019

BOReL - Results

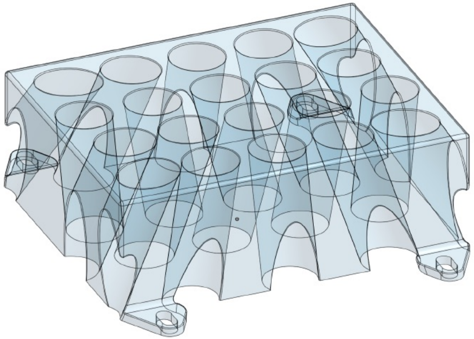






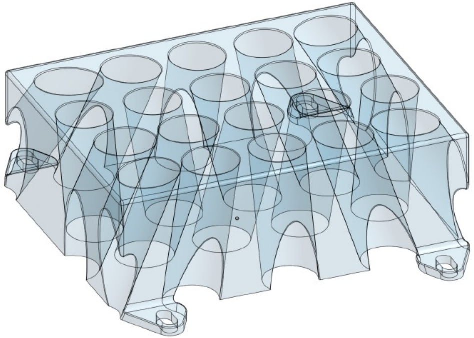
Dorfman, Shenfeld, **T.** Offline Meta RL – Identifiability Challenges and Effective Data Collection Strategies. NeurIPS 2021

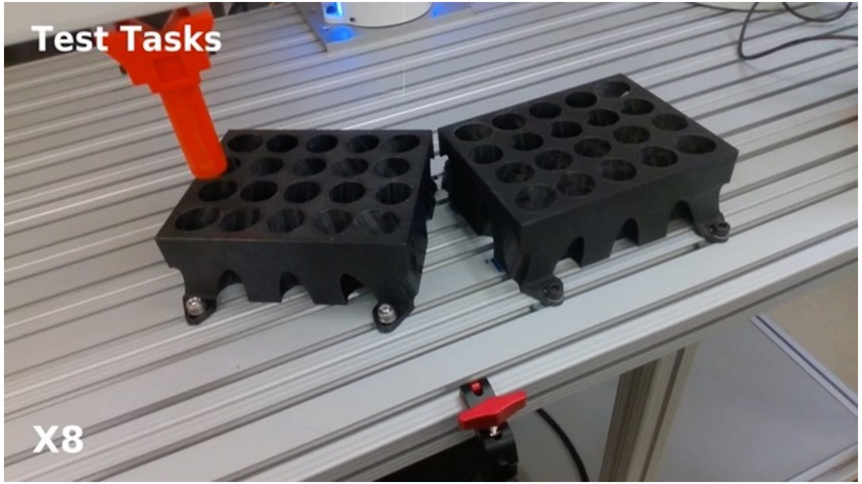
RoBOReL[1]



[1] Levy, Daniel, Jurgenson, Krupnik, Shafer, **T.** RoBOReL: Bayes Optimal Offline Meta RL for Robotic Manipulation, In Prep.

RoBOReL[1]

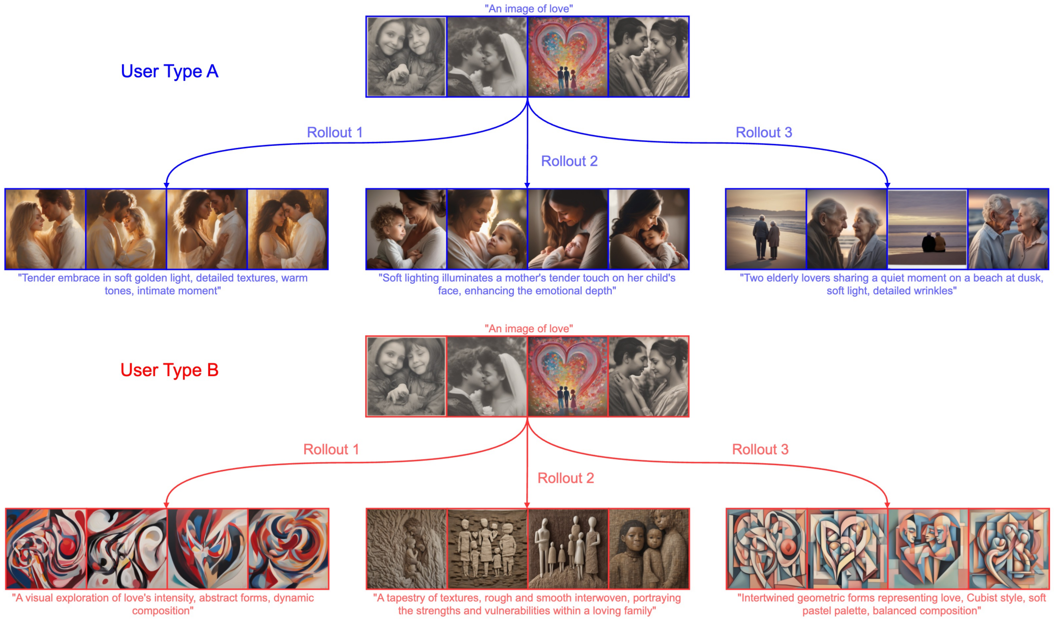




[1] Levy, Daniel, Jurgenson, Krupnik, Shafer, **T.** RoBOReL: Bayes Optimal Offline Meta RL for Robotic Manipulation, In Prep.

Additional Applications (Offline Meta RL)

- Interactive image generation[1]
- User type = MDP M



[1] Nabati et al. Preference Adaptive and Sequential Text-to-Image Generation. 2025

Generalization

Fundamental Question

$$\frac{1}{N}\sum_{i=1}^{N} \cancel{\mathbb{E}_{M\sim P(M)}} \left[\mathbb{E}_{M_i} \left[\sum_{t=0}^{H} r_t \right] \right] \Rightarrow \mathbb{E}_{M\sim P(M)} \left[\mathbb{E}_{M} \left[\sum_{t=0}^{H} r_t \right] \right]$$

N **training** tasks

$M_i \sim P(M)$

Expected **test** task performance

How many tasks N do we need to generalize?

[1] **T.**, Soudry, Zisselman. Regularization Guarantees Generalization in Bayesian Reinforcement Learning through Algorithmic Stability. AAAI 2022
[2] Rimon, **T.**, Adler. Meta Reinforcement Learning with Finite Training Tasks - a Density Estimation Approach. NeurIPS 2022.

Fundamental Question

$$\frac{1}{N}\sum_{i=1}^{N} \cancel{\mathbb{E}_{M\sim P(M)}} \left[\mathbb{E}_{M_i}\left[\sum_{t=0}^{H} r_t\right]\right] \Rightarrow \mathbb{E}_{M\sim P(M)}\left[\mathbb{E}_{M}\left[\sum_{t=0}^{H} r_t\right]\right]$$

N **training** tasks

$M_i \sim P(M)$

Expected **test** task performance

- Lower bound: $\exp(H)$
- Policy regularization →stability →generalization bounds

[1] **T.**, Soudry, Zisselman. Regularization Guarantees Generalization in Bayesian Reinforcement Learning through Algorithmic Stability. AAAI 2022

[2] Rimon, **T.**, Adler. Meta Reinforcement Learning with Finite Training Tasks - a Density Estimation Approach. NeurIPS 2022.

Parametric MDP Formulation

- A parametric space Θ and an MDP mapping $g(\theta)\colon\ \Theta \to M$
- Parameters distribution $P(\theta)$ (Lipschitz)
- Performance

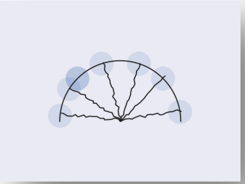
$$\mathrm{L}(\pi) = \mathbb{E}_{\theta\sim P}\left[\mathbb{E}_{\pi;\, M=g(\theta)}\left[\sum_{t=0}^{H} r(s_t, a_t)\right]\right]$$

- Bayes-optimal policy $\pi_{BO}(h_t)$

Rimon, **T.**, Adler. Meta Reinforcement Learning with Finite Training Tasks - a Density Estimation Approach. NeurIPS 2022.

The Learning Problem

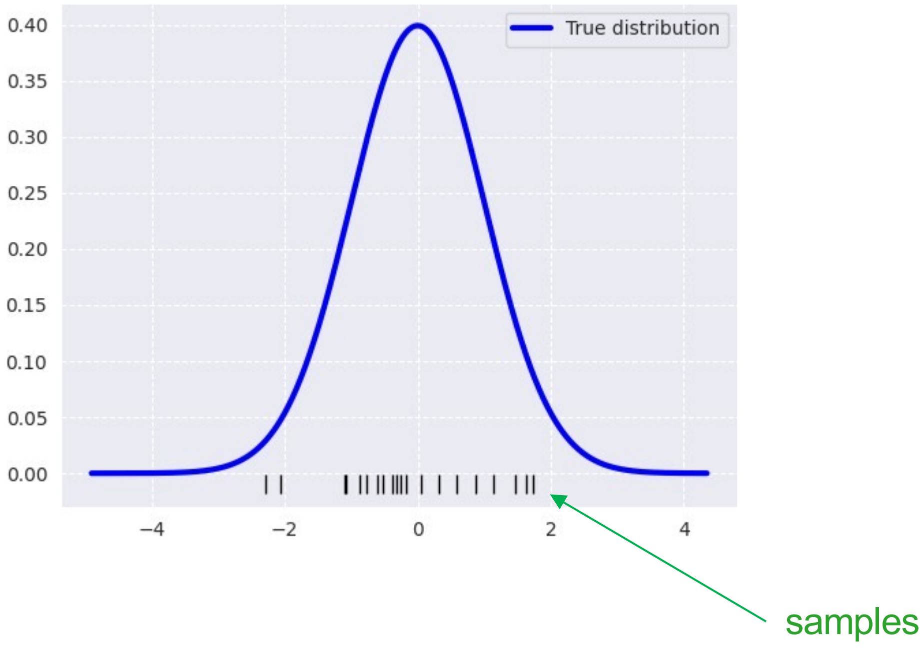
- Given N simulators $\theta_1, \dots, \theta_N \sim P(\theta)$
- Find π with low regret



$$R(\pi) = \mathrm{L}(\pi_{BO}) - \mathrm{L}(\pi)$$

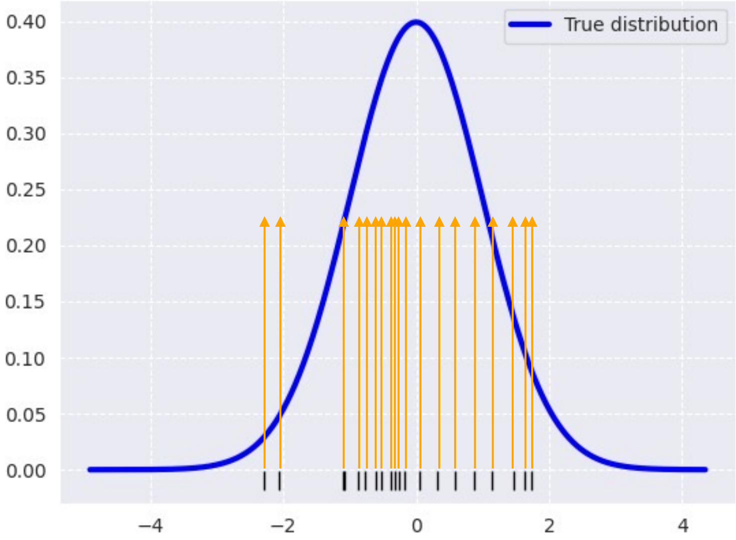
- Essentially: approximate $P(\theta)$ from samples!

Density Estimation



Density Estimation

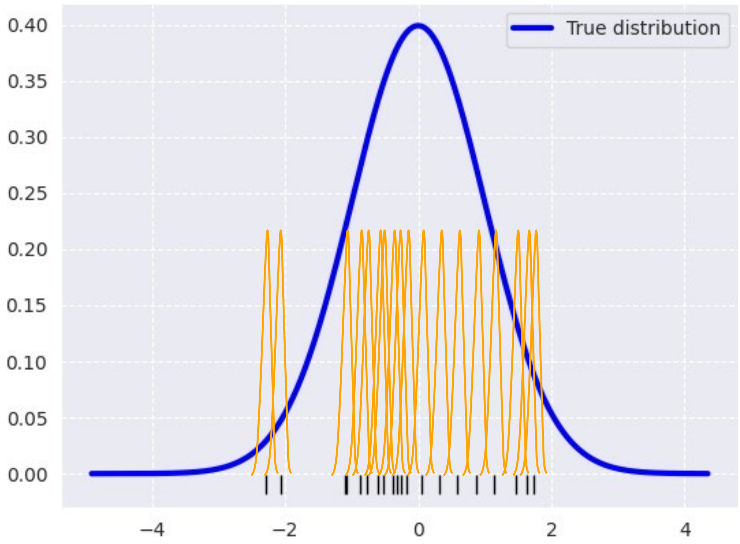
- $\hat{P}_{empirical}(\theta) = \frac{1}{N}\sum_{i=1}^{N} \delta\left(\theta - \theta_i\right)$



$$\lim_{N\to\infty} \left|P - \hat{P}_{empirical}\right|_1 \neq 0$$

Kernel Density Estimation (KDE)

- $\hat{P}_G(\theta) = \frac{1}{N}\sum_{i=1}^{N} G(\theta - \theta_i)$



Kernel Density Estimation (KDE)

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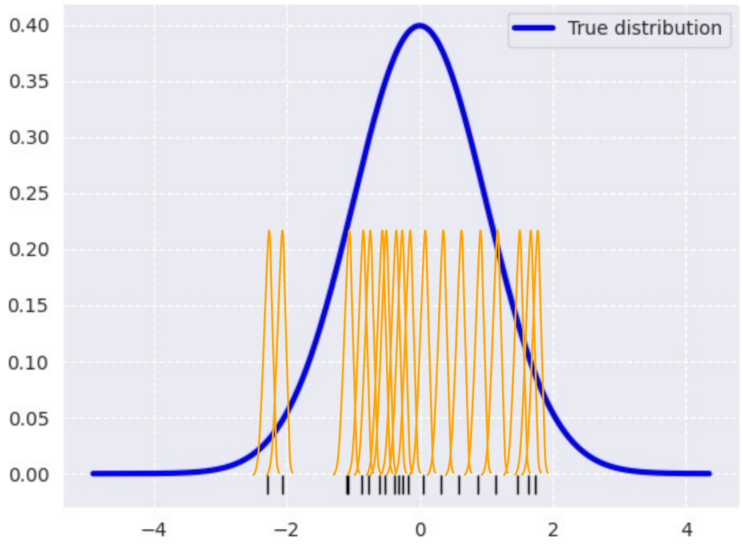
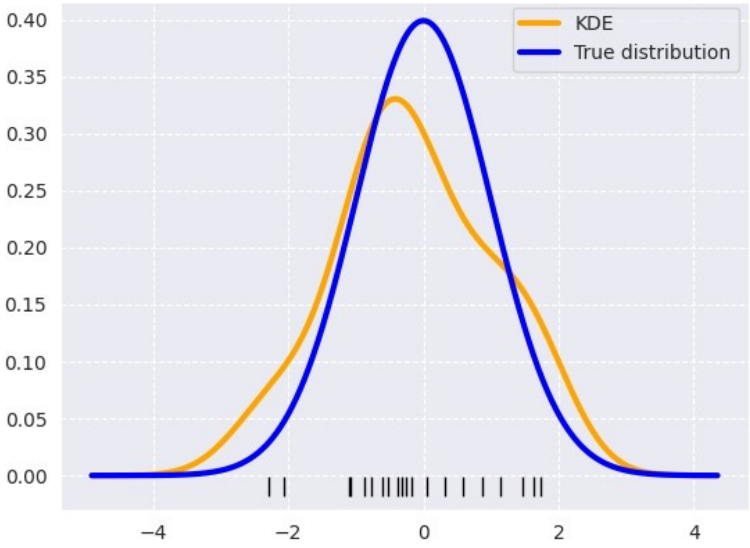




Induces regularization!

$$\lim_{N\to\infty} \left| P - \hat{P}_G(\theta) \right|_1 = 0$$

PAC Bounds

- $\mathrm{L}(\pi) = \mathbb{E}_{\theta \sim \hat{P}_G(\theta)} \left[\mathbb{E}_{\pi;\, M=g(\theta)} \left[\sum_{t=0}^{H} r(s_t, a_t) \right] \right] \Rightarrow \hat{\pi}_G^*$

PAC Bounds

- $L(\pi) = \mathbb{E}_{\theta \sim \hat{P}_G(\theta)}\left[\mathbb{E}_{\pi;\, M=g(\theta)}\left[\sum_{t=0}^{H} r(s_t, a_t)\right]\right] \Rightarrow \hat{\pi}_G^*$

dim(Θ)

- W.P. $1 - 1/N$: $R(\hat{\pi}_G^*) \leq 2C_{max}H|\Theta|C_d\left(\frac{\log(N)}{N}\right)^{\frac{1}{2+d}}$

poly(d)

PAC Bounds

- $\mathrm{L}(\pi) = \mathbb{E}_{\theta \sim \hat{P}_G(\theta)} \left[\mathbb{E}_{\pi;\, M=g(\theta)} [\sum_{t=0}^{T} r(s_t, a_t)] \right] \Rightarrow \hat{\pi}_G^*$

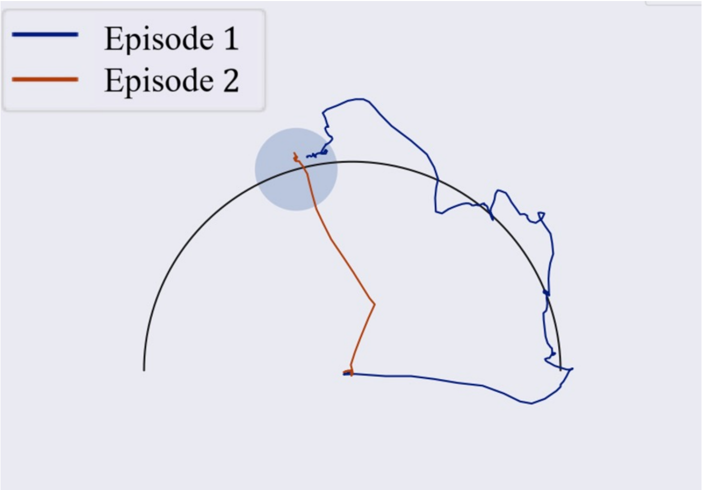
$\dim(\Theta)$

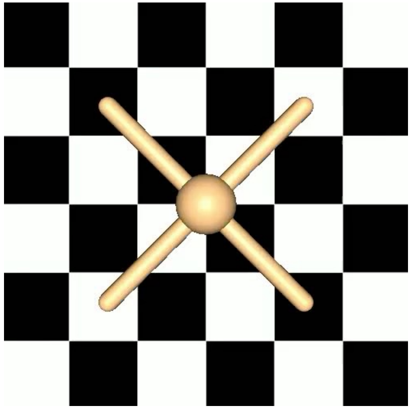
- W.P. $1 - 1/N$: $R(\hat{\pi}_G^*) \le 2 C_{max} H |\Theta| C_d \left(\frac{\log(N)}{N} \right)^{\frac{1}{2+d}}$

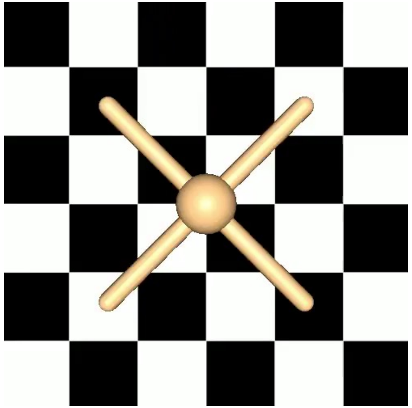
- Proof idea: KDE bounds + simulation lemma

No direct dependency on $|S|, |A|$
Linear growth with H
Exponential in d

Theory vs. Practice







$$\boldsymbol{dim(\Theta_L) = 1}$$

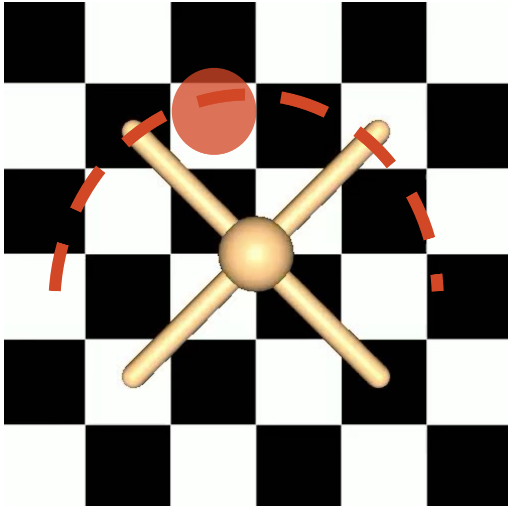
$$T \approx 100$$

$$N \approx 40$$

Task Space Dimensionality

Easy Problems

- Low task dim[1]



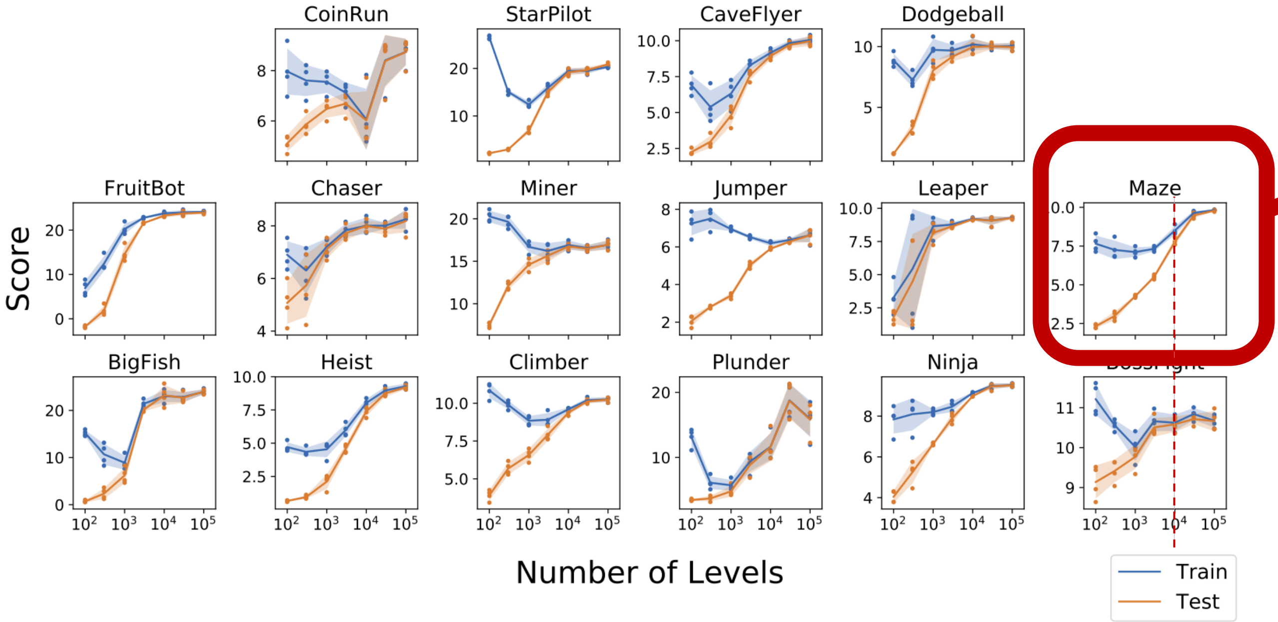
Hard Problems

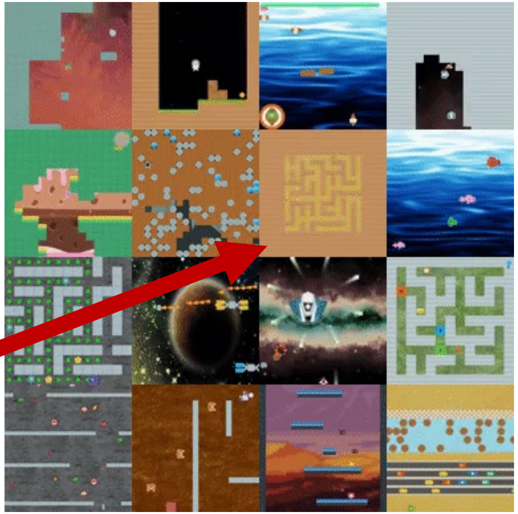
- High task dim[2]

[1] Rimon, **T.**, Adler. Meta Reinforcement Learning with Finite Training Tasks - a Density Estimation Approach. NeurIPS 2022.
[2] Mandi et al. On the Effectiveness of Fine-tuning Versus Meta-reinforcement Learning. 2022

Zero Shot Generalization

- ProcGEN benchmark – still a challenge



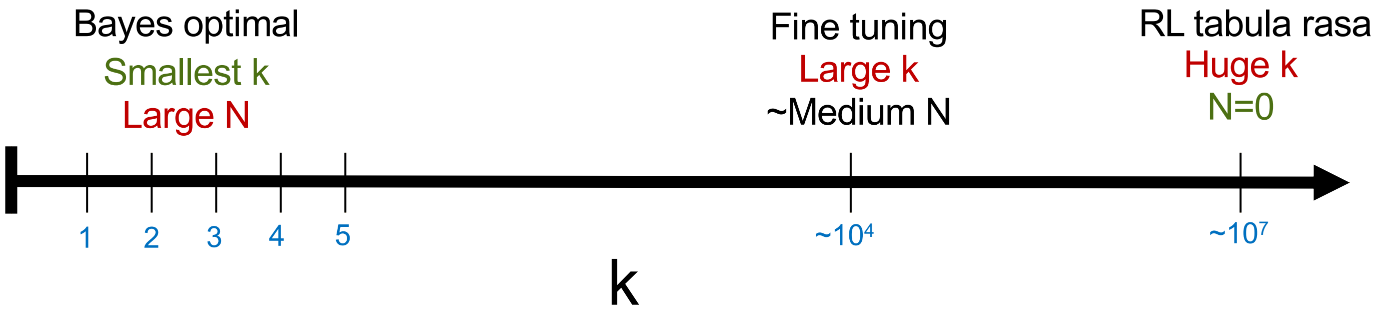


Cobbe et al. Leveraging procedural generation to benchmark reinforcement learning, 2020

Adaptation-Generalization Tradeoff

- k-shot learning in test task
- N training tasks

Goal: $\max_{\pi} \mathbb{E}_{M\sim P(M)}\left[\mathbb{E}_M\left[\sum_{t=0}^{H=kT} r_t\right]\right]$



Adaptation-Generalization Tradeoff

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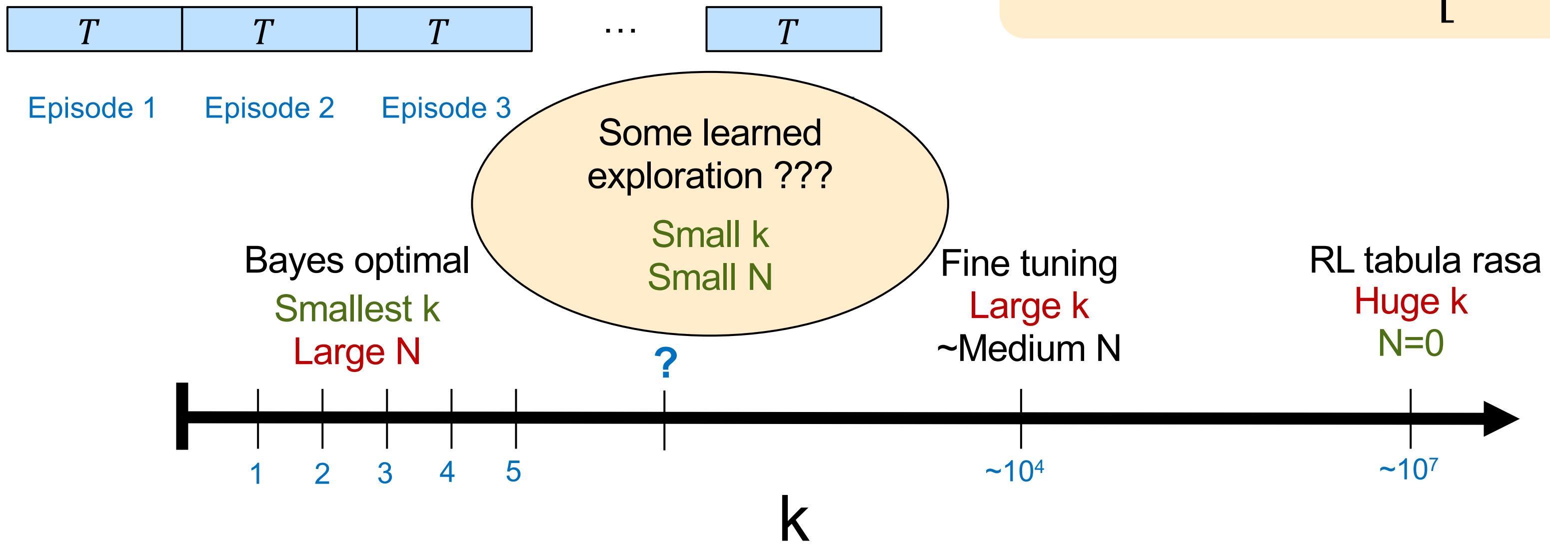
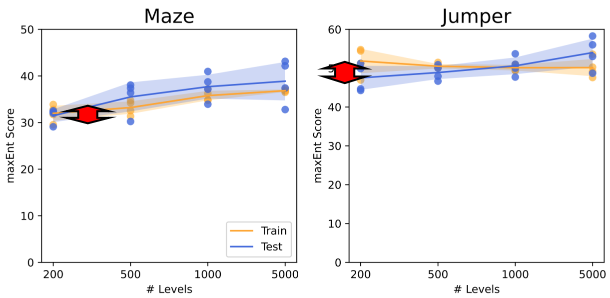


A. Tamar, Technion

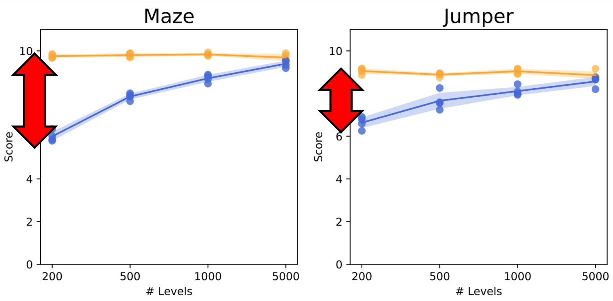
Explore to Generalize

- **Observation** – Max Entropy[1] exploration **generalizes**[2]!

Max Entropy



Max Reward

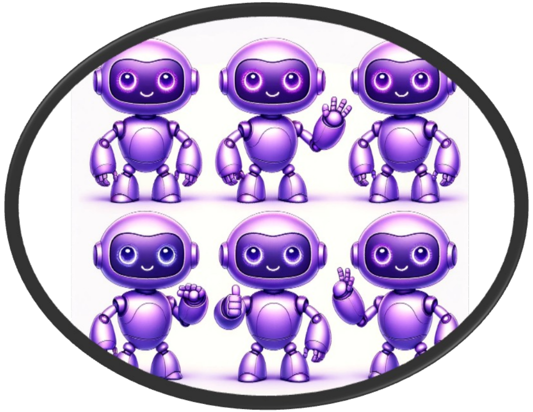


Why? Exploration behavior harder to memorize!

[1] Mutti et al. The importance of non-markovianity in maximum state entropy exploration. 2022
[2] Zisselman, Lavie, Soudry, **T.** Explore to Generalize in Zero-Shot RL. NeurIPS 2023

Explore to Generalize

- Exp-Gen Algorithm

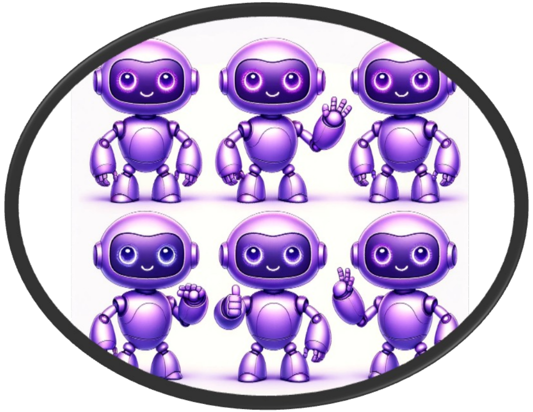


Max-Reward
Ensemble

Max-Entropy
Agent

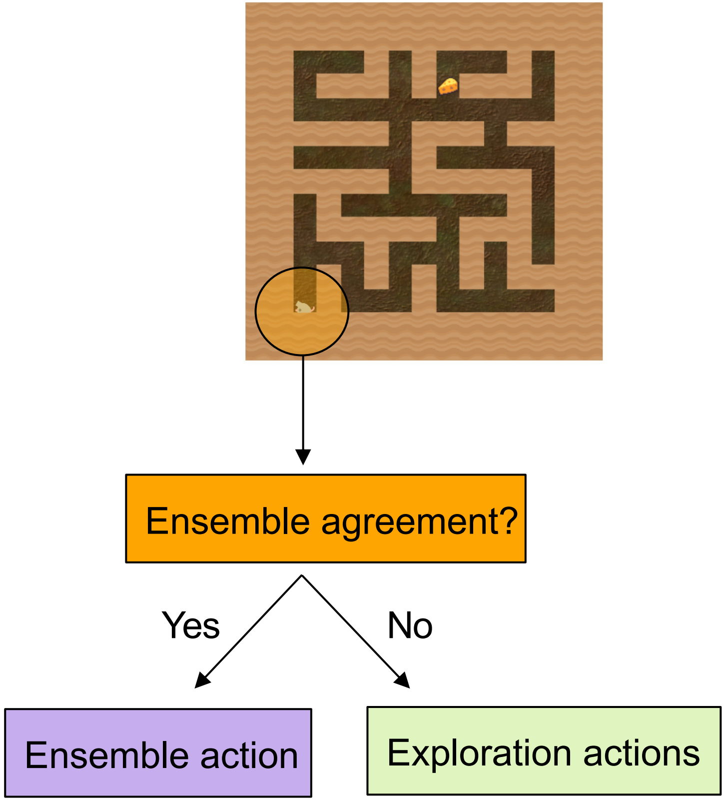
Explore to Generalize

- Exp-Gen Algorithm



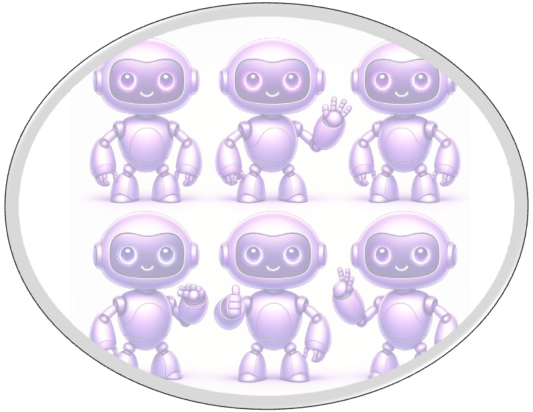
Max-Reward Ensemble

Max-Entropy Agent



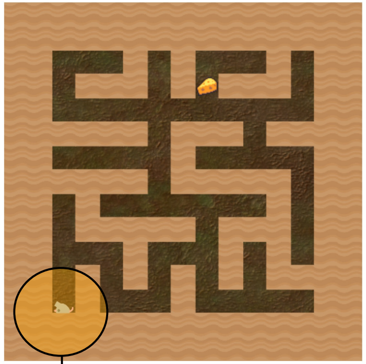
Explore to Generalize

- Exp-Gen Algorithm



Max-Reward
Ensemble

Max-
Agent



Generalize
→Exploit

Ensemble agreement?

Yes

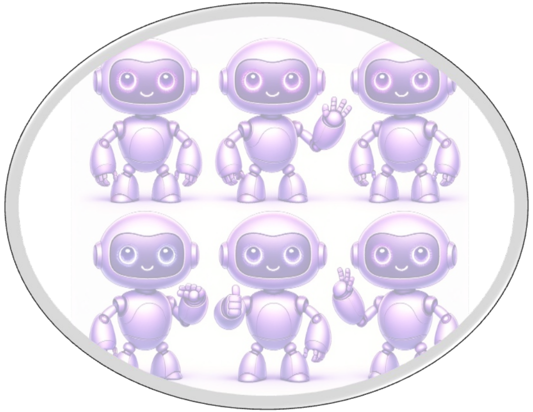
No

Ensemble action

Exploration actions

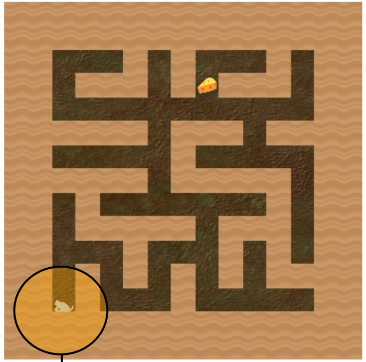
Explore to Generalize

- Exp-Gen Algorithm



Max-Reward
Ensemble

Max-
Agent



Generalize
→Exploit

Ensemble agreement?

Yes

No

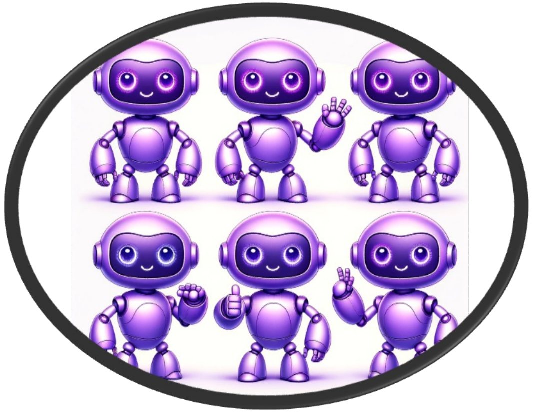
Generalize
→Explore

Ensemble action

Exploration actions

Explore to Generalize

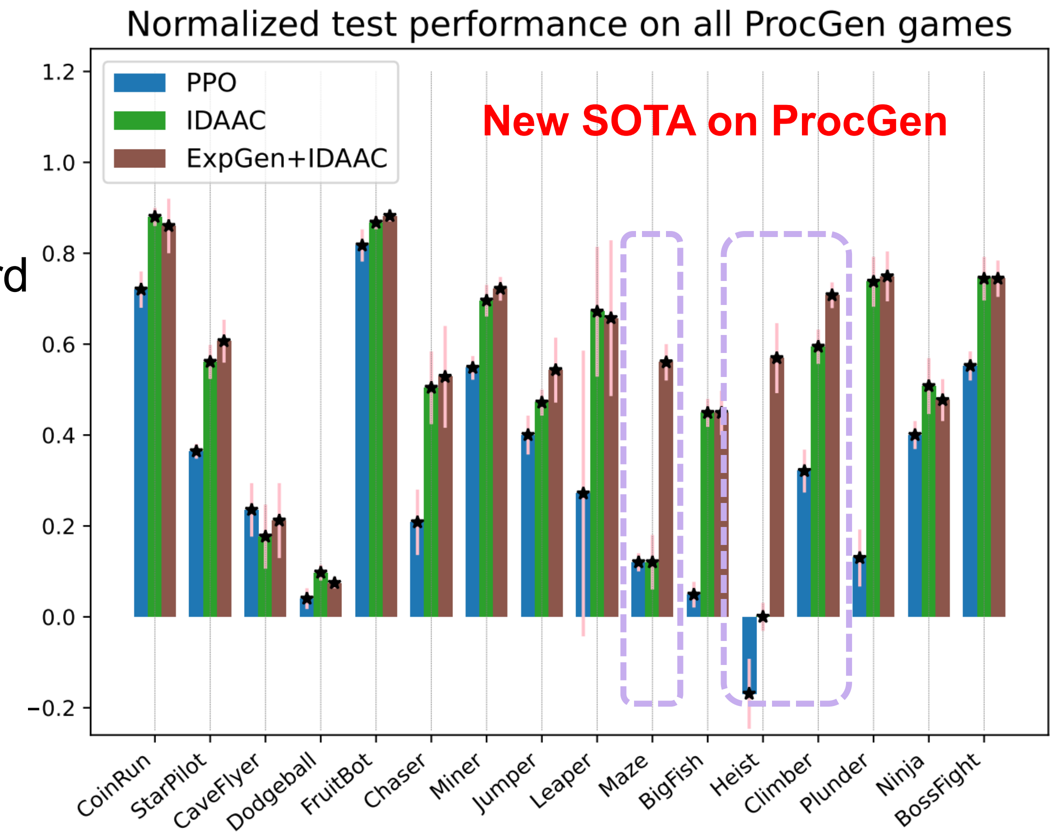
- Exp-Gen Algorithm



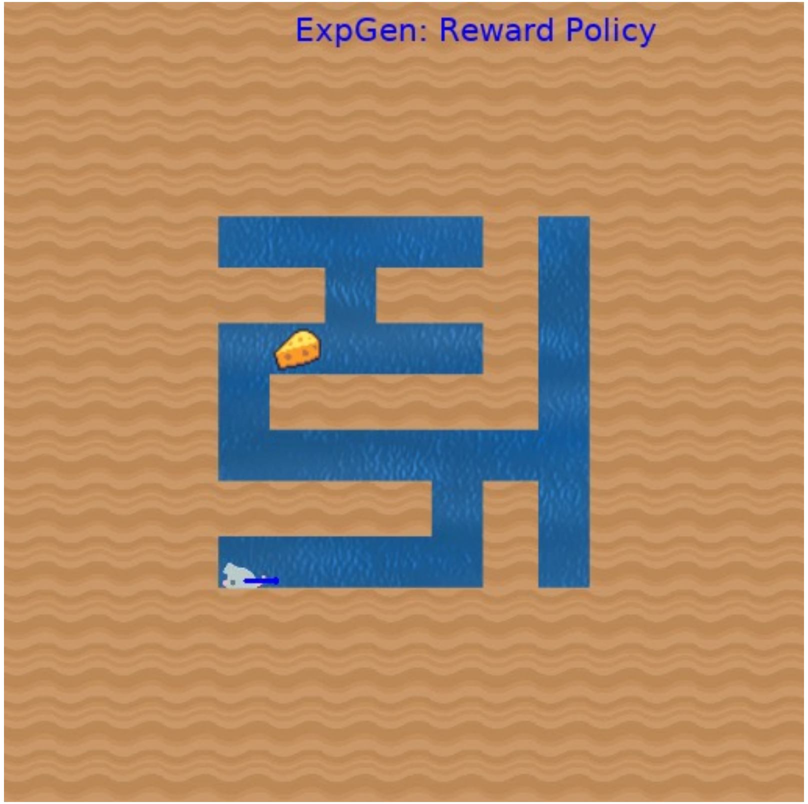
Max-Reward Ensemble

IDAAC

Max-Entropy Agent



Explore to Generalize



Fundamental Question

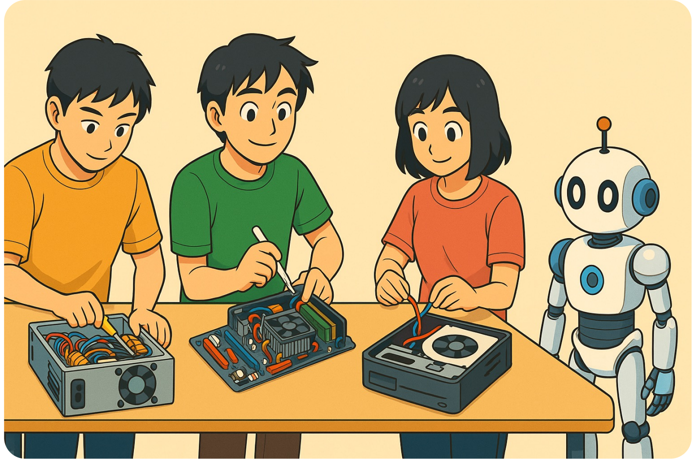
A fundamental connection between exploration and generalization?

[1] Zisselman, Mutti, Francis-Meretzki, Shafer, **T.** Blindfolded Experts Generalize Better: Insights from Robotic Manipulation and Videogames, NeurIPS 2025

Imitation Learning

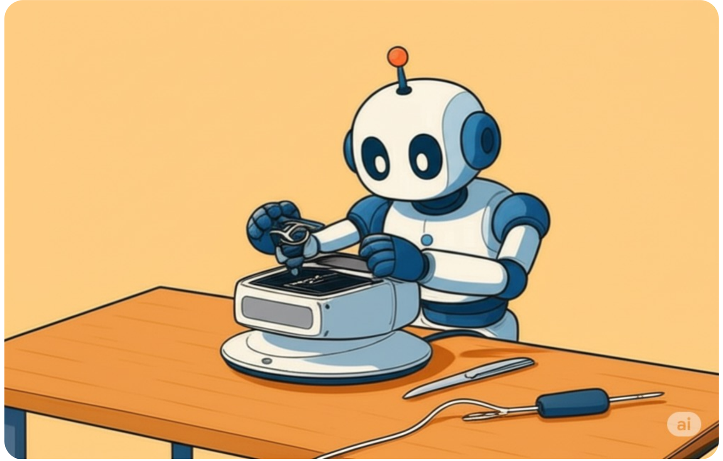
Train

Demonstrations of a small set of tasks



Test

Zero-shot generalization on unseen tasks

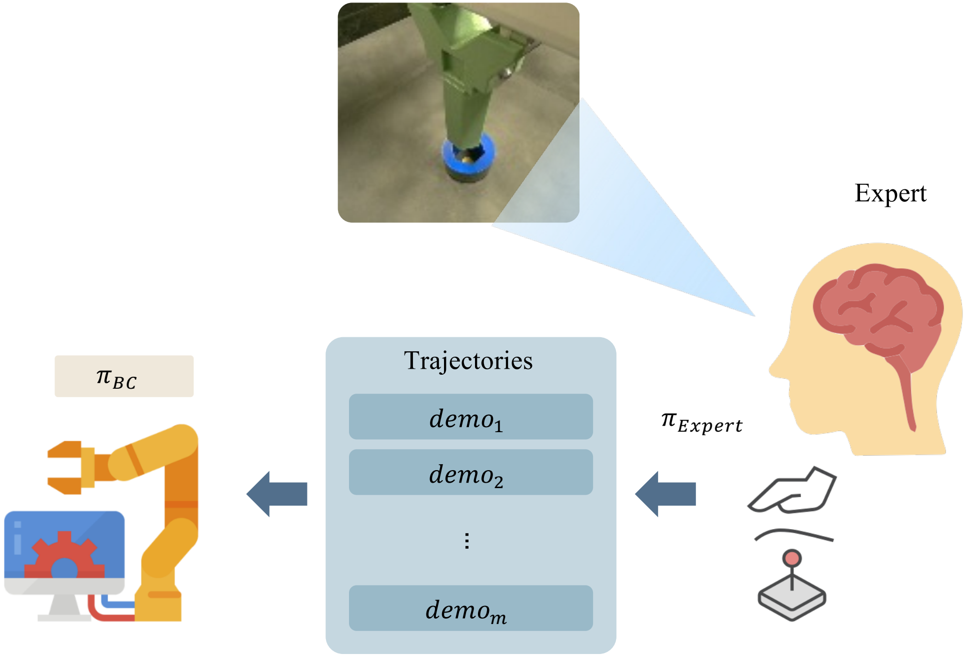


Basically, what everyone in robotics industry is doing these days…

Main question:
Can we **guide** experts' behavior toward better **generalization**?

Exploration → Generalization

<u>Idea:</u> Force experts to **explore** by **reducing task information**!

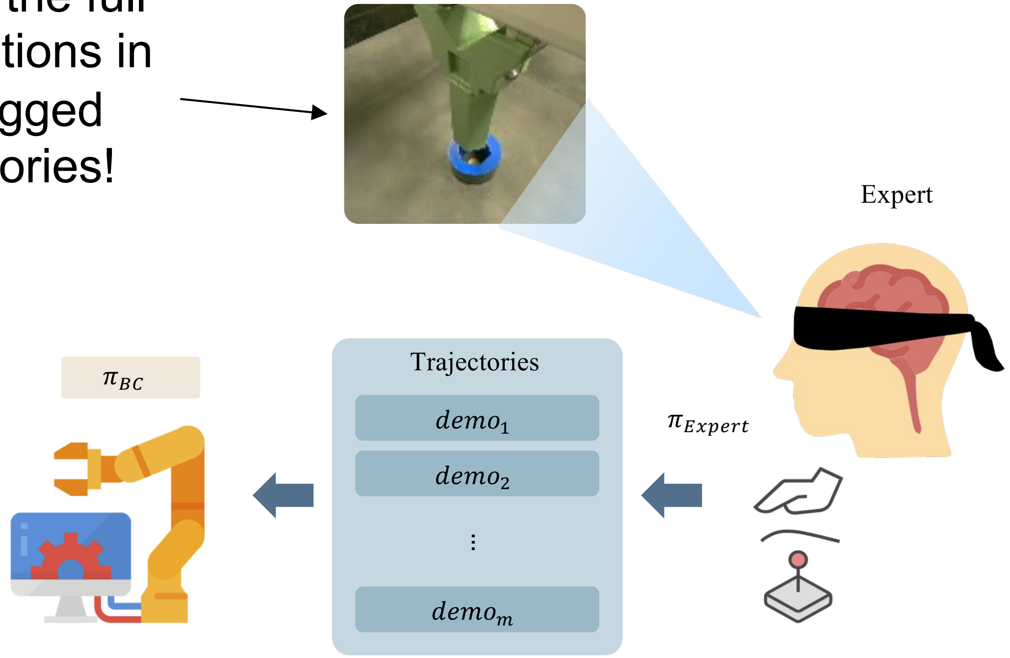


How to reduce information?

Blindfold the experts!

Record the full observations in the logged trajectories!

put a blindfold

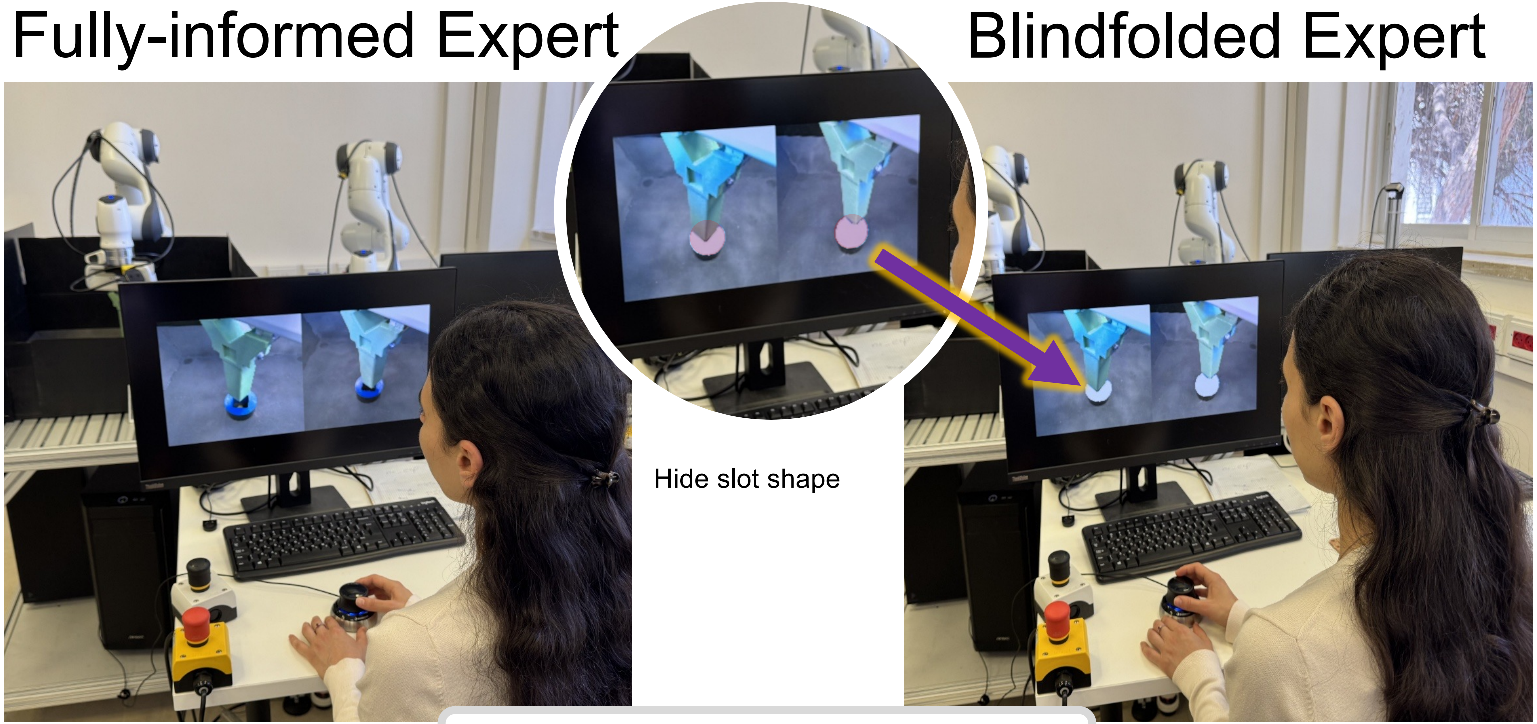


How to reduce information?

How to apply the blindfold? – mask observations!

Fully-informed Expert

Blindfolded Expert



Hide slot shape

Record the full observations!

Theory

Setting: m tasks, each with n demonstrated trajectories

$$error_{gen} \lesssim O\left(\frac{1}{n}\right) + O\left(\frac{I_{Z;T}}{m}\right) + \varepsilon_{gen}(\pi^E) + \varepsilon_{opt}(\hat{\pi})$$

Theory

<u>Setting</u>: m tasks, each with n demonstrated trajectories

$$error_{gen} \lesssim O\left(\frac{1}{n}\right) + O\left(\frac{I_{Z;T}}{m}\right) + \varepsilon_{gen}(\pi^E) + \varepsilon_{opt}(\hat{\pi})$$

Theory

<u>Setting</u>: m tasks, each with n demonstrated trajectories

We focus on this!

Task information available to the expert

$$error_{gen} \lesssim O\left(\frac{1}{n}\right) + O\left(\frac{\boxed{I_{Z;T}}}{m}\right) + \varepsilon_{gen}(\pi^E) + \varepsilon_{opt}(\hat{\pi})$$

Expert suboptimality

Optimization error

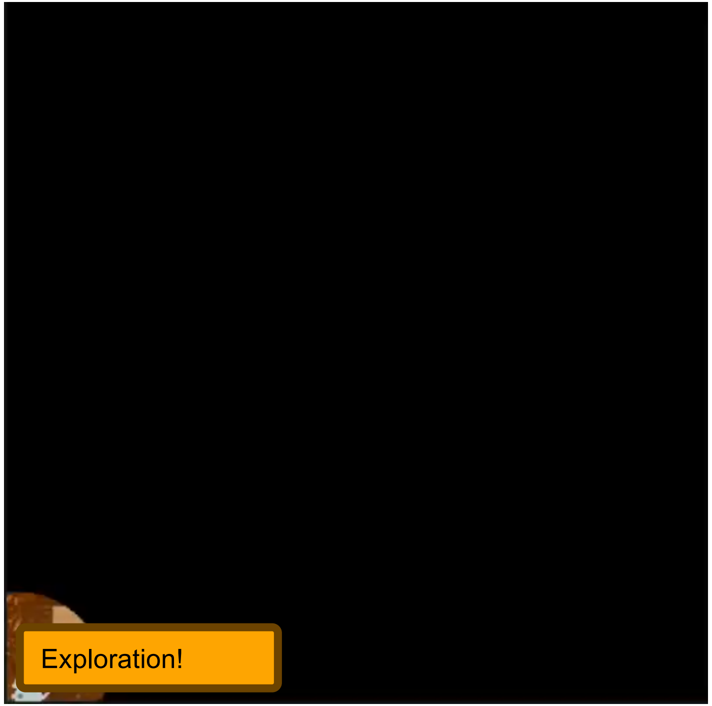
Bounded [1]
Zero for success/fail tasks

[1] Chen, Hu, Jin, Li, Wang. Understanding domain randomization for sim-to-real transfer. ICLR 2022.

Expert Observation – ProcGen Maze

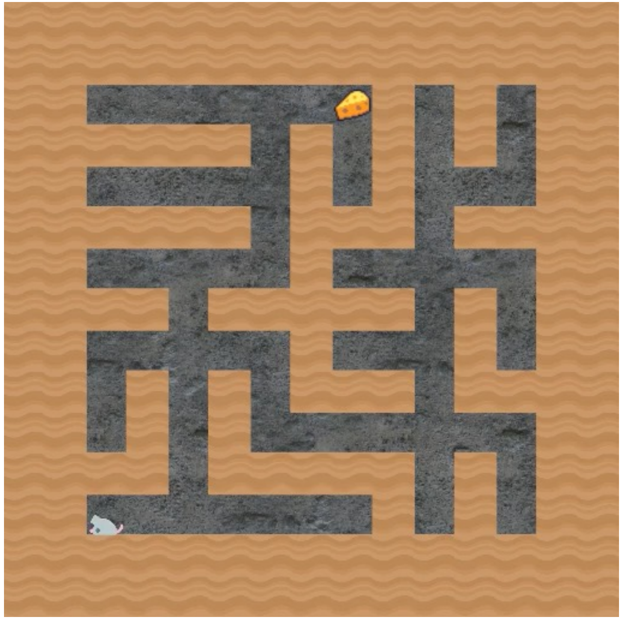
Fully-informed Expert

Blindfolded Expert



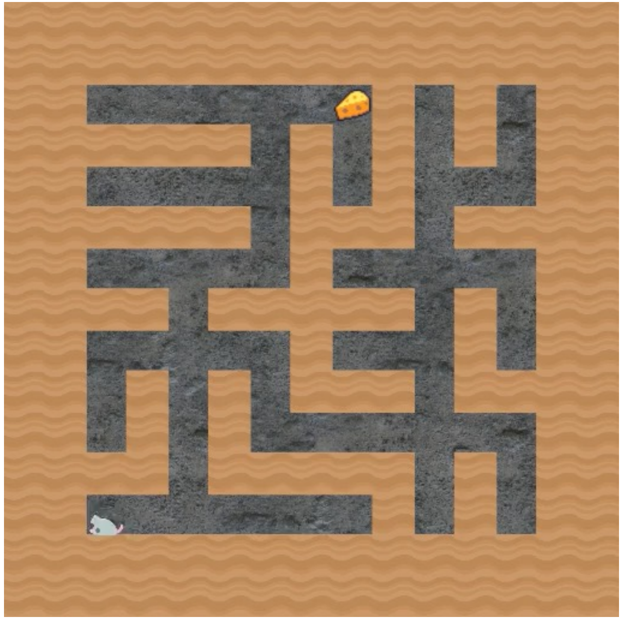
Test Results - ProcGen Maze

Fully-informed Expert



Lost in the maze

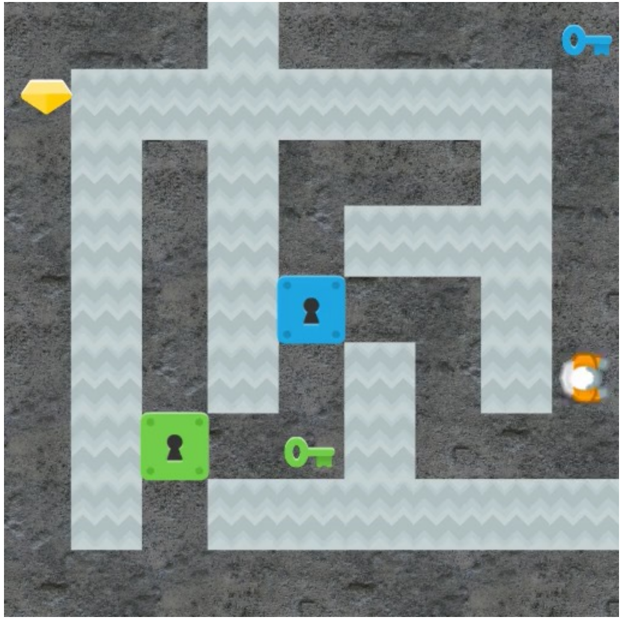
Blindfolded Expert



Explore the maze

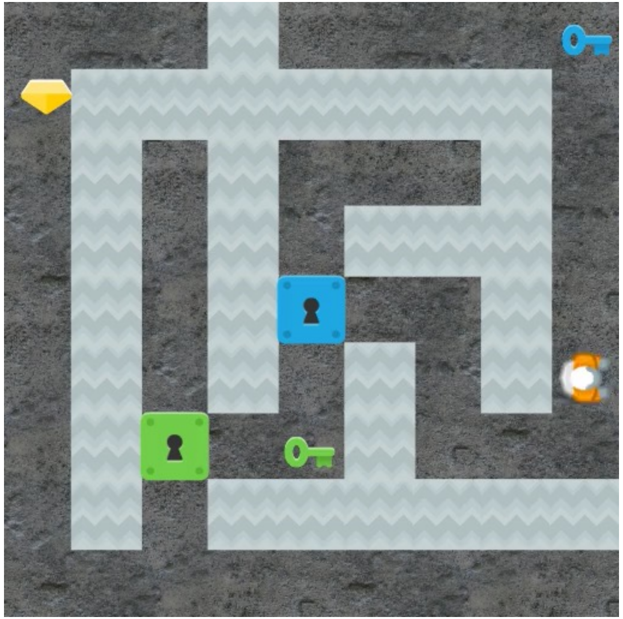
Test results – ProcGen Heist

Fully-informed Expert



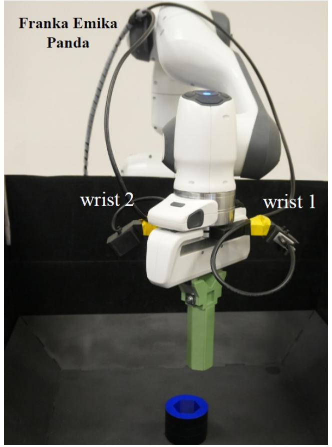
Lost in the maze

Blindfolded Expert

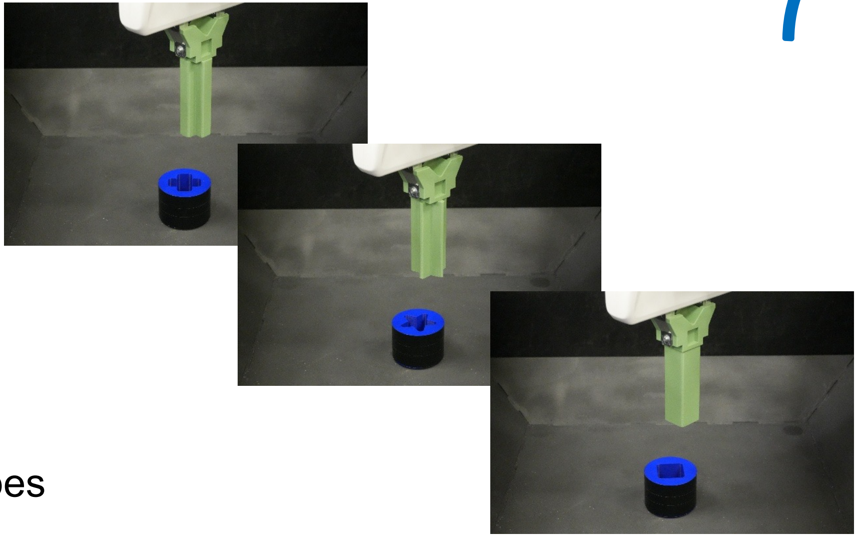


Explore the maze

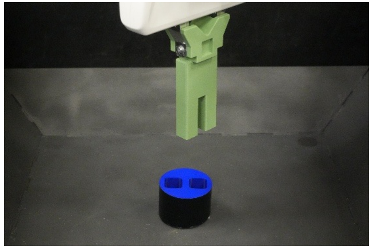
Robot experiment – Peg insertion



Train



Test

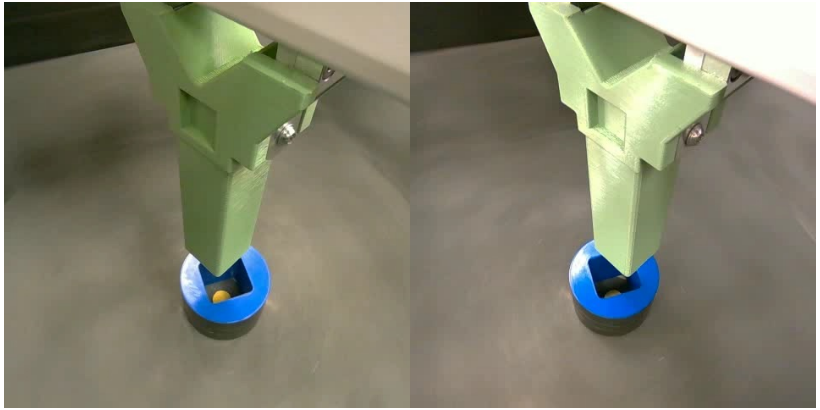


- Adapted from FMB [1]
- **Train** on only 2-5 shapes
- **Goal**: insert the peg
- **Generalization is very difficult** → Agents memorize training shapes

[1] Luo, Xu, Liu, Tan, Lin, Wu, Abbeel, Levine. Fmb: a functional manipulation benchmark for generalizable robotic learning. IJRR 2025.

Expert Observation – Peg Insertion

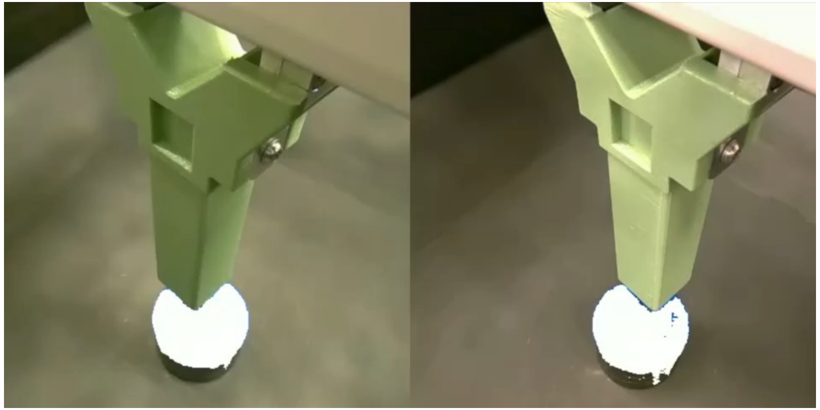
Fully-informed Expert



Normal speed

Align → Insert

Blindfolded Expert

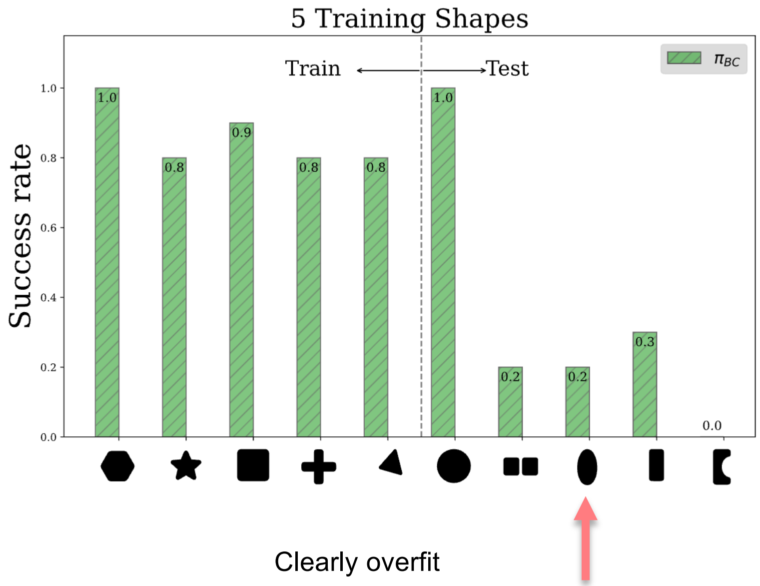


Speedup x8

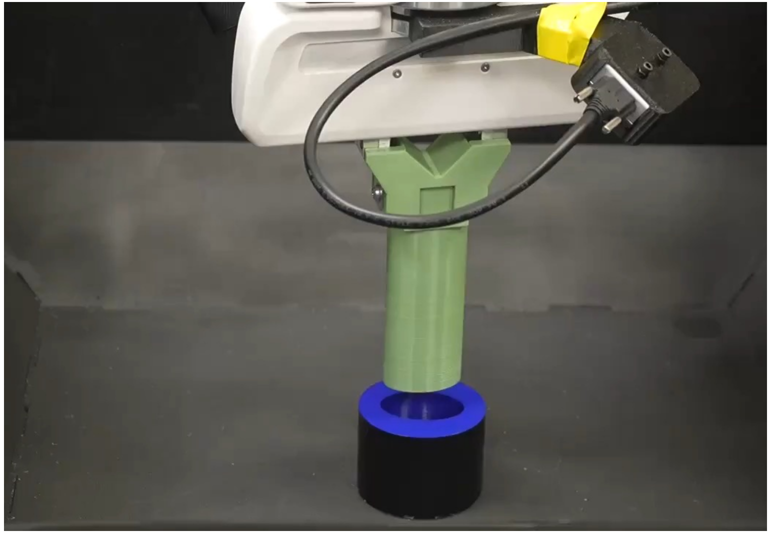
Explore → Insert

Generalization Results – Peg Insertion

Test shape – Fully-informed Expert (Failure)

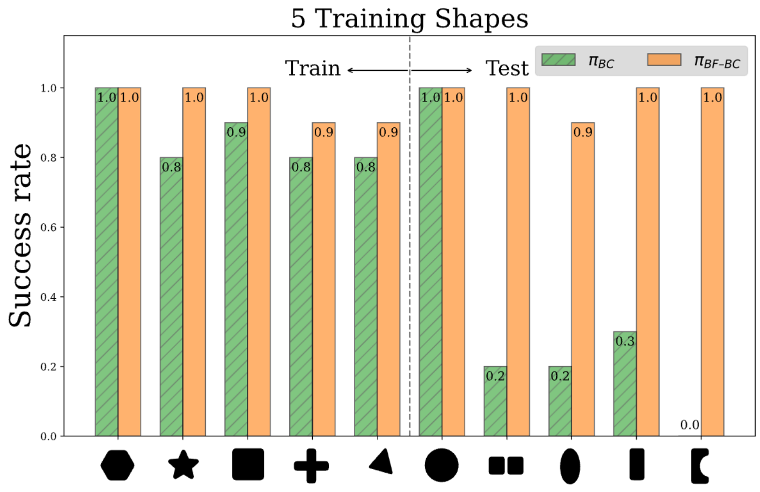


Clearly overfit

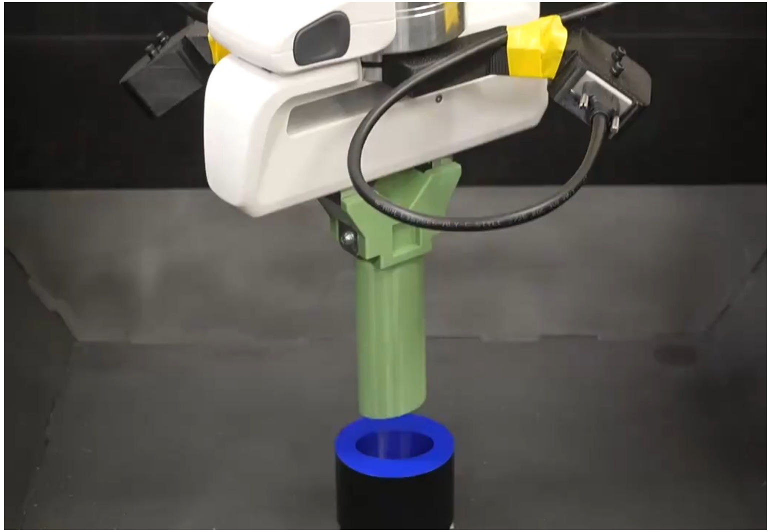


Generalization Results – Peg Insertion

Test shape – Blindfolded Expert (Success)



Generalization achieved



Summary and Outlook

- RL in real world must adapt, generalize
- World models → POMDP information states
- Adaptation-Exploration tradeoff

Future research

- Compositional generalization
  - Language-based[1]
  - Object-centric models[2]
- Other data
  - Foundation models (LLMs, World models)

Thanks to students and collaborators!

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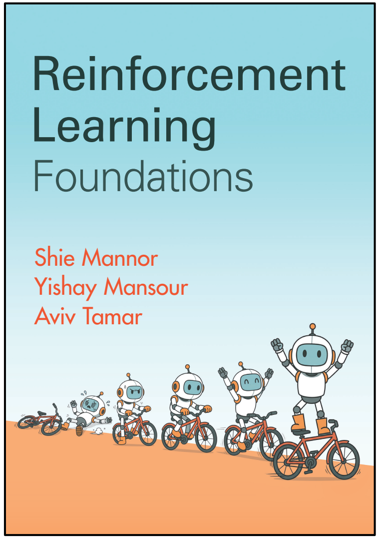
Orr Krupnik
Itay Lavie
Adi Levy
Mirco Mutti
Zohar Rimon
Eli Shafer
Idan Shenfeld
Daniel Soudry
Ev Zisselman

[1] E. Jang. To Understand Language is to Understand Generalization. 2021
[2] Haramati, Daniel, **T.** Entity-centric reinforcement learning for object manipulation from pixels. 2024

Learn More

- Latest research updates
- Links to papers/code
- **New book**
  - **Cambridge University Press 2026**



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